

INTERNSHIP REPORT ON BACKEND WEB DEVELOPMENT USING JAVASCRIPT FRAMEWORKS AT DIGICON TECHNOLOGIES PLC

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Submitted By—

Shahriar Hasan

Student ID: 2002138

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Industrial Internship Certificate



Digicon Technologies PLC

242/A Tejgaon I/A, Gulshan Link Road, Dhaka-1208.
Phone: +88 09610 110 110
Email : info@digicontechnologies.com
Web : www.digicontechnologies.com

Ref: DTPLC/HR/2026/01996
Date: September 08, 2026

TO WHOM IT MAY CONCERN

This is to certify that Mr. Shahriar Hasan, student of BSC in Electronics and Communication Engineering, Hajee Mohammad Danesh Science and Technology University has successfully completed an internship program from August 16, 2026 to September 05, 2026 at Information Technology Department in Digicon Technologies PLC. During the period of his internship, he was enthusiastic, hardworking and passionate towards his work.

We wish him all the best for his future endeavors.

For Digicon Technologies PLC

A handwritten signature in black ink, appearing to be "Sajib", written over a horizontal line.

Md. Saiful Islam Sajib
Manager | Human Resources



Acknowledgements

First and foremost, all praises and profound gratitude are due to Almighty Allah, the Most Merciful and Most Beneficent, who bestowed upon me the health, intellectual resilience, and perseverance required to complete this industrial internship and compile this comprehensive report.

I express my deepest gratitude, profound respect, and indebtedness to my respected **Academic Supervisor** in the Department of Electronics and Communication Engineering at Hajee Mohammad Danesh Science and Technology University (HSTU). His critical evaluations, scholarly feedback, and constructive academic stewardship were instrumental in contextualizing my industrial learnings within the foundational principles of computer science and communication engineering.

I extend my heartfelt thanks to the Chairman and all distinguished faculty members of the Department of Electronics and Communication Engineering, HSTU, for providing a vibrant academic ecosystem, rigorous theoretical foundations, and continuous moral support throughout my undergraduate studies.

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I would also like to thank the engineering team members, DevOps specialists, and Quality Assurance engineers at Digicon for their collegial camaraderie, technical discussions during daily Scrum stand-ups, and collaborative spirit.

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Shahriar Hasan
Student ID: 2002138
HSTU, Dinajpur

October, 2026

Executive Summary

This internship report provides a comprehensive, rigorous synthesis of an industrial training attachment completed at **Digicon Technologies PLC**, Dhaka, Bangladesh, as a mandatory curricular requirement for the degree of Bachelor of Science in Electronics and Communication Engineering at Hajee Mohammad Danesh Science and Technology University (HSTU). Digicon Technologies PLC is one of Bangladesh's premier Business Process Outsourcing (BPO) and Information Technology Enabled Services (ITES) providers, operating at an enterprise scale with over 1,500 Full-Time Equivalents (FTEs) and processing more than 3 million multi-channel customer transactions monthly across telecommunications, banking, fintech, healthcare, and government sectors.

Embedded as a **Software Engineering Intern** within the Software Engineering and Technology Solutions Division, the author's primary mandate centered on backend system architecture, API engineering, and distributed service optimization utilizing modern JavaScript and TypeScript frameworks. The core technologies employed include the **Node.js** asynchronous V8 runtime, **Express.js**, and **NestJS** frameworks, backed by polyglot persistence infrastructure comprising **PostgreSQL** for relational consistency, **MongoDB** for flexible ticket documents, and **Redis** for distributed in-memory caching and session management.

During the three-month tenure, the author actively contributed to two mission-critical production platforms:

1. **CRM Support Ticket Lifecycle & Event Routing Engine:** Engineered high-throughput RESTful endpoints using the Controller-Service-Repository layered architectural pattern. Implemented dynamic Service Level Agreement (SLA) breach monitoring, optimistic concurrency control to prevent race conditions during ticket assignment, and WebSocket event propagation for real-time contact center agent workspace synchronization.
2. **Enterprise SMS Gateway & Asynchronous Dispatch Pipeline:** Architected a resilient messaging microservice designed to decouple inbound client HTTP requests from downstream telecommunication carrier networks. The pipeline incorporates a **BullMQ / Redis** message queue, a **Token Bucket** rate-limiting algorithm enforcing telecom compliance (1,500 SMS/second threshold), an exponential backoff retry mechanism for transient network failures, and an **HMAC-SHA256** cryptographic signature verifier for inbound delivery receipt (DLR) webhooks.

To ensure production robustness, security hardening was executed using **Helmet.js**,

Cross-Origin Resource Sharing (CORS) whitelisting, and Role-Based Access Control (RBAC) via JSON Web Tokens (JWT). Automated unit and integration testing pipelines were established using **Jest** and **Supertest**, attaining upwards of 85% code coverage across core service layers. Furthermore, applications were packaged into lightweight production containers utilizing multi-stage **Docker** builds, reducing final image sizes by 83% (from 840 MB to 142 MB). Performance profiling with **Autocannon** confirmed that the integrated Redis caching layer reduced database query latency by approximately 60%, scaling sustained throughput from 1,220 to over 9,200 requests per second under concurrent loads.

This report documents the organizational landscape, technical foundations, architectural blueprints, concrete code implementations, DevOps pipelines, real-world engineering challenges, and qualitative professional reflections gained during this industrial immersion. The experience solidified the practical translation of university computing theory into scalable, fault-tolerant enterprise software engineering.

Keywords: Backend Web Development, Node.js, Express.js, NestJS, Digicon Technologies, BPO, CRM, SMS Gateway, Redis Caching, PostgreSQL, MongoDB, RESTful API, Docker, RBAC, Microservices Architecture.

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List of Acronyms and Abbreviations

Acronym	Full Description
ACID	Atomicity, Consistency, Isolation, Durability
API	Application Programming Interface
BACCO	Bangladesh Association of Contact Center & Outsourcing
BPO	Business Process Outsourcing
BTRC	Bangladesh Telecommunication Regulatory Commission
CORS	Cross-Origin Resource Sharing
CPU	Central Processing Unit
CRM	Customer Relationship Management
CRUD	Create, Read, Update, Delete
CSAT	Customer Satisfaction Score
DAO	Data Access Object
DBMS	Database Management System
DLR	Delivery Receipt (Telecommunication Messaging)
DNS	Domain Name System
DTO	Data Transfer Object
ECE	Electronics and Communication Engineering
ERP	Enterprise Resource Planning
FTE	Full-Time Equivalent (Workforce Measurement)
HMAC	Hash-based Message Authentication Code
HRMS	Human Resource Management System
HSTU	Hajee Mohammad Danesh Science and Technology University
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
I/O	Input / Output
IETF	Internet Engineering Task Force
IP	Internet Protocol
IT	Information Technology
ITES	Information Technology Enabled Services
IVR	Interactive Voice Response
JSON	JavaScript Object Notation
JWT	JSON Web Token
MVC	Model-View-Controller
NoSQL	Not Only SQL (Non-relational distributed database)
ODM	Object-Document Mapper
ORM	Object-Relational Mapper
OWASP	Open Web Application Security Project
PR	Pull Request (Version Control)
QA	Quality Assurance
RDBMS	Relational Database Management System
RBAC	Role-Based Access Control

Acronym	Full Description
REST	Representational State Transfer
RFC	Request for Comments (Internet Standard)
SDK	Software Development Kit
SLA	Service Level Agreement
SMPP	Short Message Peer-to-Peer Protocol
SMS	Short Message Service
SOC	Security Operations Center
SQL	Structured Query Language
TCP	Transmission Control Protocol
TPS	Transactions / Telegrams Per Second
TTL	Time To Live (Cache Expiration)
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
UUID	Universally Unique Identifier
V8	Google Open-Source High-Performance JavaScript & WebAssembly Engine
VCS	Version Control System

Chapter 1

Introduction

1.1 Background and Context of Industrial Attachment

In the contemporary computing and communication engineering landscape, the boundary separating theoretical academia from commercial software engineering has become increasingly dynamic. While undergraduate university curricula provide foundational mathematical rigor, algorithmic thinking, database theory, and operating system principles, industrial software development requires mastery over distributed systems, high-concurrency microservices, cloud-native deployments, and strict operational Service Level Agreements (SLAs).

To bridge this quintessential gap, the Bachelor of Science in Electronics and Communication Engineering curriculum at Hajee Mohammad Danesh Science and Technology University (HSTU) incorporates a mandatory credit-bearing **Industrial Training / Internship (Course Code: ECE 450)**. This requirement affords senior-year undergraduate students an immersive opportunity to apply classroom concepts directly within leading enterprise environments, understand corporate workflows, and contribute to production-grade software engineering pipelines.

The author was afforded the opportunity to complete this industrial attachment at **Digicon Technologies PLC**, one of the vanguard Information Technology and Business Process Outsourcing (BPO) enterprises headquartered in Dhaka, Bangladesh. Founded in 2010, Digicon operates at a significant national and international scale, employing over 1,500 Full-Time Equivalents (FTEs) across state-of-the-art contact center facilities and software development units, processing over 3 million multi-channel customer interactions monthly [1]. The company provides a diversified portfolio encompassing voice and non-voice BPO operations, custom enterprise SaaS software, telecommunication integration services, and human resource management systems.

Within this organizational matrix, the author was positioned as a **Software Engineering Intern** within the Software Engineering and Technology Solutions Division. The focus of the internship centered on **Backend Web Development** utilizing modern JavaScript and TypeScript frameworks, including **Node.js**, **Express.js**, and **NestJS**, backed by polyglot database clusters comprising **PostgreSQL**, **MongoDB**, and **Redis**. This monograph presents an exhaustive, academically rigorous documentation of the architectural, technical, operational, and personal learnings garnered throughout the three-month industrial engagement.

1.2 Significance of the Internship in the BSc Curriculum

The field of Electronics and Communication Engineering (ECE) has evolved far beyond physical-layer signal processing and basic embedded microcontrollers. Modern communication engineering is inherently intertwined with software-defined networking, cloud computing, distributed microservice architectures, and high-throughput application programming interfaces (APIs) [2]. The significance of this internship attachment in the author's undergraduate curriculum is multifaceted:

1. **Synthesizing Theory into Practice:** While university laboratory experiments typically feature isolated, single-user desktop applications, enterprise web development requires architecting solutions that scale to thousands of concurrent users, adhere to strict latency constraints, and ensure uninterrupted 24/7/365 availability.
2. **Mastery of the Asynchronous Event-Driven Paradigm:** Traditional procedural programming models do not naturally translate to high-concurrency network servers. Working with the Node.js V8 runtime allowed the author to practically understand non-blocking I/O, event loops, thread-pool offloading via Libuv, and reactive programming [3].
3. **Understanding Enterprise Polyglot Persistence:** Rather than utilizing a single relational database for all application concerns, industrial systems employ specialized databases according to data velocity and structural characteristics. The internship offered direct experience in synchronizing relational data (PostgreSQL) with document-oriented schemaless entities (MongoDB) and in-memory key-value caching (Redis) [4].
4. **Exposure to Industrial DevOps and Quality Assurance:** Modern software delivery relies on Continuous Integration and Continuous Deployment (CI/CD) pipelines, automated testing pyramids (Unit, Integration, End-to-End), and containerization via Docker [5]. Experiencing these pipelines firsthand within production workflows provides critical competencies that complement undergraduate coursework.
5. **Cultivation of Professional and Soft Skills:** Navigating complex enterprise requirements demands effective communication, adherence to Agile/Scrum sprint ceremonies, cross-functional collaboration with Frontend and QA teams, and disciplined Git version control etiquette.

1.3 Purpose and Specific Objectives of the Training

The overarching purpose of the industrial training was to gain end-to-end practical competency in designing, implementing, securing, and maintaining scalable backend web services

for enterprise applications. To realize this purpose, specific technical and professional objectives were formulated at the onset of the attachment:

- **Objective 1: In-depth Understanding of Enterprise Host Operations:** Investigate and document the operational, technical, and commercial ecosystem of Digicon Technologies PLC, with particular emphasis on how its contact center BPO operations integrate with internal software solutions.
- **Objective 2: Architectural Design of Scalable Microservices:** Formulate and analyze multi-tiered backend system architectures adhering to the Controller-Service-Repository pattern, ensuring separation of concerns, code maintainability, and horizontal scalability.
- **Objective 3: Core Backend Service Implementation:** Implement production-grade RESTful APIs for two flagship internal modules:
 - A *CRM Support Ticket Lifecycle & Event Routing Engine* capable of automated SLA monitoring, agent assignment, and real-time WebSocket state synchronization.
 - An *Enterprise SMS Gateway Microservice* featuring asynchronous message queuing (BullMQ/Redis), token bucket rate-limiting algorithms, and cryptographic HMAC-SHA256 webhook signature verification.
- **Objective 4: Security and Access Control Hardening:** Implement defense-in-depth security mechanisms, including JSON Web Token (JWT) authentication, Role-Based Access Control (RBAC) authorization guards, password hashing using Bcrypt, and HTTP header security through Helmet.js in alignment with OWASP recommendations [6].
- **Objective 5: Quality Assurance and Containerization:** Construct automated unit and integration test suites using Jest and Supertest achieving at least 80% code coverage; containerize microservices using lightweight multi-stage Dockerfiles.
- **Objective 6: Performance Optimization and Benchmarking:** Conduct empirical performance profiling using Autocannon and k6 to evaluate response latency and throughput gains achieved through multi-tier Redis caching under varying concurrent loads.

1.4 Scope and Limitations of the Report

1.4.1 Scope of Work

The technical scope of this report encompasses the design, implementation, testing, and deployment of backend services constructed during the internship period from July 01, 2026 to September 30, 2026. Specifically, it covers:

- Architectural analysis and schema design for the Customer Relationship Management (CRM) and SMS Gateway backend subsystems.
- Code-level exposition of routing controllers, business service layers, data access repositories, and custom validation pipelines.
- Security configurations including JWT token lifecycle management, RBAC authorization, and API rate-limiting middleware.
- Automated testing pipelines, Docker containerization scripts, and benchmarking methodology.

1.4.2 Operational Limitations and Confidentiality

In accordance with the corporate Non-Disclosure Agreement (NDA) executed between the author and Digicon Technologies PLC, certain proprietary parameters have been purposefully abstracted or sanitized:

- Proprietary customer personally identifiable information (PII), live telephone numbers, internal corporate passwords, and secret cryptographic keys have been fully sanitized, anonymized, or replaced with representative synthetic values.
- Commercial client contracts, telecommunication pricing tariffs, and confidential financial metrics are omitted from this academic report.
- Code samples presented in this monograph are focused on architectural paradigms, structural logic, and design patterns rather than proprietary commercial trade secrets.

1.5 Report Outline and Organizational Structure

This monograph is structured into eight thematic chapters, methodically organized to take the reader from foundational enterprise context to deep technical implementation, empirical evaluation, and future recommendations:

- **Chapter 1: Introduction** establishes the academic context of the internship, highlights its significance within the HSTU BSc curriculum, details primary objectives, and defines the scope and limitations of the study.

- **Chapter 2: Host Organization Profile — Digicon Technologies PLC** provides an extensive overview of the host company, its history, corporate vision, business divisions (BPO and Tech solutions), organizational hierarchy, software development lifecycle, and the engineering division structure.
- **Chapter 3: Technical Background & Technology Stack** presents a comprehensive technical review of the JavaScript and TypeScript backend ecosystem, dissecting the Node.js V8 event-loop runtime, Express.js and NestJS frameworks, TypeScript type safety, REST architectural constraints, PostgreSQL, MongoDB, Redis caching, and web security standards.
- **Chapter 4: System Architecture & Requirements Analysis** details the functional and non-functional requirements, presents the overall multi-tier layered microservices blueprint, database Entity-Relationship schemas, CRM ticket event flows, SMS gateway asynchronous pipelines, and DevOps architectures.
- **Chapter 5: Design & Implementation of Backend Services** offers an in-depth code-level walkthrough of the implemented systems, including authentication and RBAC guards, CRM ticket management controllers, the BullMQ-driven SMS dispatch queue with Token Bucket throttling, Redis caching implementations, and global error handling filters.
- **Chapter 6: Testing, Quality Assurance & DevOps** documents the testing strategy, unit test coverage with Jest, integration testing with Supertest, multi-stage Docker containerization, and empirical performance benchmarking comparing direct database queries against Redis caching under stress.
- **Chapter 7: Professional Experience, Challenges & Work Environment** reflects upon the workplace culture at Digicon, Agile/Scrum sprint ceremonies, Git collaboration etiquette, real-world technical bottlenecks faced and their engineering resolutions, and personal soft skill acquisition.
- **Chapter 8: Conclusion & Future Recommendations** synthesizes the key achievements of the internship, evaluates the alignment with academic learning, offers strategic architectural recommendations for Digicon's engineering platforms, and provides personal reflections on prospective career trajectories.

Chapter 2

Host Organization Profile: Digicon Technologies PLC

2.1 Corporate History and Genesis

Digicon Technologies PLC (formerly Digicon Technologies Ltd.) stands as one of the pioneer and premier Information Technology and Business Process Outsourcing (BPO) conglomerates in Bangladesh. Established in **2010**, the company was founded with a strategic vision to position Bangladesh as a competitive global destination for IT-enabled services (ITES) and sophisticated enterprise software engineering. Over the past decade and a half, Digicon has undergone exponential institutional expansion, evolving from a boutique contact center into a diversified, multi-vertical technology corporation [1].

Headquartered strategically at the Rajuk Trade Center in Nikunja-2, Khilkhet, Dhaka, with secondary operational hubs and data facilities situated across Mirpur and Tejgaon, Digicon boasts over 100,000 square feet of state-of-the-art infrastructural space. The organization is a recognized founding and corporate member of the **Bangladesh Association of Contact Center & Outsourcing (BACCO)** and maintains strong affiliations with the Bangladesh Computer Samity (BCS) and the Bangladesh Association of Software and Information Services (BASIS) [7].

Throughout its operational history, Digicon has forged strategic partnerships with multinational corporations, domestic telecom giants (including Grameenphone, Robi Axiata, Banglalink, and Teletalk), leading private commercial banks, non-banking financial institutions, multinational pharmaceutical enterprises, and various governmental emergency helplines. The company's consistent adherence to international quality benchmarks is evidenced by its ISO 9001 (Quality Management System) and ISO 27001 (Information Security Management System) certifications, ensuring rigorous data governance and operational integrity across all technical deliveries.

2.2 Corporate Vision, Mission, and Core Values

The operational ethos of Digicon Technologies PLC is anchored upon clear corporate philosophies that drive its engineering methodologies and business customer engagements.

2.2.1 Corporate Vision

“To be the preeminent technology and outsourcing partner in South Asia, empowering global enterprises through strategic digital transformation, automated operational excellence, and state-of-the-art software solutions.”

2.2.2 Corporate Mission

Digicon translates its corporate vision into actionable engineering and service commitments:

- **Technological Innovation:** Continuously engineer robust, scalable, and intuitive software platforms that solve acute enterprise bottlenecks across Customer Relationship Management, messaging, and human capital administration.
- **Operational Excellence:** Maintain uninterrupted 99.98% uptime across mission-critical customer support services and backend telecommunication gateways.
- **Human Capital Empowerment:** Foster an ecosystem of continuous technical upskilling, mentorship, and professional growth for youth and engineering graduates in Bangladesh.
- **Value Creation:** Deliver measurable Return on Investment (ROI) and cost efficiencies for enterprise partners through agile software automation and intelligent resource routing.

2.2.3 Core Values

The corporate and engineering culture at Digicon is underpinned by four core institutional tenets:

1. **Integrity and Confidentiality:** Unwavering adherence to ethical data governance, customer confidentiality, and non-disclosure standards.
2. **Agility and Adaptability:** Rapid responsiveness to changing technological landscapes, evolving client specifications, and emergent software design paradigms.
3. **Customer Centricity:** Aligning backend architecture and system response times directly with user experience and business customer satisfaction metrics.
4. **Collaborative Synergy:** Fostering cross-disciplinary teamwork between software developers, DevOps practitioners, quality assurance testers, and domain business analysts.

2.3 Business Domains and Operational Verticals

Digicon Technologies operates across three primary, mutually reinforcing business pillars: Business Process Outsourcing (BPO) and Contact Center Operations, Software and Technology Solutions, and Professional Training and IT Consultancy.

2.3.1 Business Process Outsourcing (BPO) & Contact Center Solutions

As a cornerstone of its commercial success, Digicon operates one of the largest contact center infrastructures in Bangladesh:

- **Workforce Scale:** Digicon manages over **1,500 Full-Time Equivalents (FTEs)** operating in 24/7/365 rotational shifts.
- **Transaction Volume:** The contact center ecosystem processes over **3 million customer transactions monthly** and manages a daily peak call volume exceeding 1.2 million interactions.
- **Multi-Channel Coverage:** The platform bridges inbound/outbound voice telephony, interactive voice response (IVR), web chat widgets, official social media messaging, and automated ticketing.
- **Global Outreach:** Services are delivered across 14 operational hubs, supporting clients spanning 40+ countries and handling interactions in multiple regional and international languages.

2.3.2 Software and Technology Solutions Division

Complementing its service operations, Digicon maintains an extensive internal software engineering wing dedicated to designing custom enterprise software, cloud-native SaaS platforms, and automation tools. The software solutions portfolio includes:

- **Customer Relationship Management (CRM):** A proprietary omnichannel CRM tailored for high-volume contact centers. It provides intelligent ticket routing, real-time agent presence monitoring, automated SLA escalation, and multi-tier customer interaction history.
- **Enterprise SMS & Telecommunication Gateway:** A high-throughput messaging microservice platform engineered to dispatch transactional notifications, OTP authentication codes, and promotional broadcasts across Bangladesh's telecommunication carriers with sub-second delivery latencies.
- **Human Resource Management System (HRMS):** An integrated enterprise system handling employee attendance, biometric verification, multi-tier leave approval hierarchies, and payroll processing.
- **Enterprise Resource Planning (ERP):** Modular software platforms supporting supply chain management, inventory tracking, field service management, and utility billing for commercial enterprises.

- **Artificial Intelligence & Conversational Agents:** Natural Language Processing (NLP) chatbots integrated into client web portals to automate tier-1 customer inquiries and deflect repetitive call center volumes.
- **Security Operations Center (SOC) & Infrastructure Services:** Continuous vulnerability management, network packet inspection, and database security auditing.

2.3.3 Professional Corporate Training and IT Consultancy

Digicon provides specialized workforce upskilling programs for corporate entities, preparing technical and non-technical staff in customer interaction standards, cyber hygiene, and modern software tooling. In addition, the firm offers strategic IT consultancy, assisting legacy enterprises in executing seamless migrations to cloud architectures and digital workflows.

2.4 Corporate and Engineering Organizational Hierarchy

The institutional structure of Digicon Technologies PLC is designed to ensure strict executive accountability, fluid cross-departmental communication, and agile technical delivery. Figure 2.1 illustrates the corporate hierarchy and the detailed structure of the Software Engineering Division.

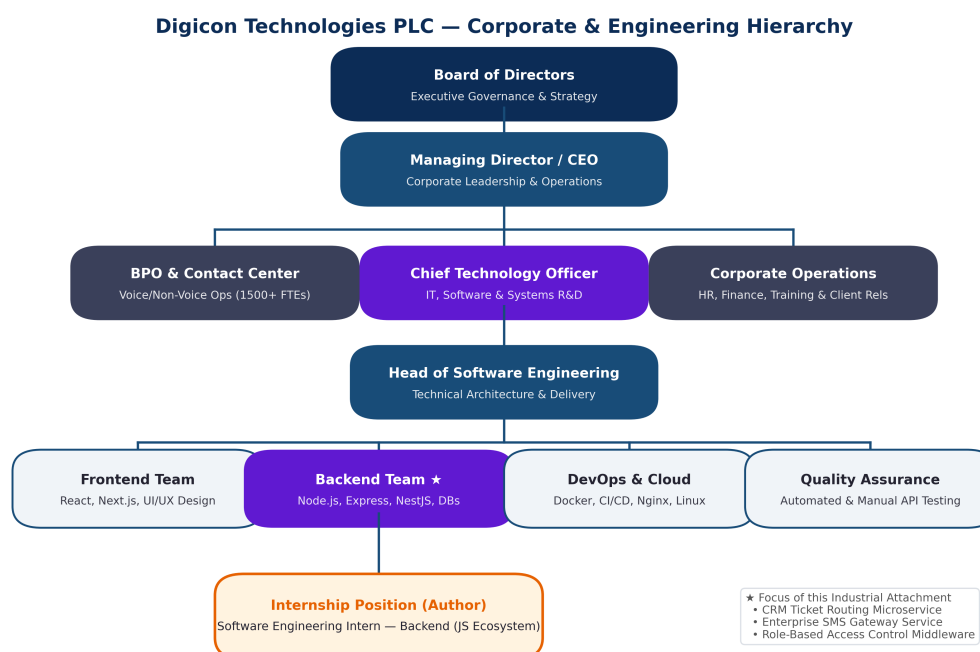


Figure 2.1: Corporate and Engineering Organizational Hierarchy of Digicon Technologies PLC.

At the apex of the corporate structure, the **Board of Directors** exercises strategic governance, capital allocation, and regulatory compliance. The **Managing Director / Chief**

Executive Officer (CEO) oversees overall enterprise performance and commercial growth. Operational execution is partitioned into three key executive wings:

- **BPO Operations Division:** Led by the Head of BPO, managing workforce logistics, contact center quality assurance, and voice operations.
- **Corporate Administration:** Managing human resources, finance, corporate client partnerships, and legal affairs.
- **Technology and Software Division:** Headed by the **Chief Technology Officer (CTO)**, who steers organizational technology strategy, infrastructure architecture, and R&D investments.

Under the CTO, the **Head of Software Engineering** directs the day-to-day software development lifecycle. The software engineering division is structured into specialized functional units:

1. **Frontend Engineering Team:** Focuses on responsive user interfaces, design systems, and client-side applications built with React, Next.js, and modern CSS frameworks.
2. **Backend Engineering Team (Internship Attachment):** Responsible for system architecture, RESTful and GraphQL API design, database modeling, business logic implementation, caching layers, and external service integrations using Node.js, Express, NestJS, PostgreSQL, MongoDB, and Redis.
3. **DevOps & Cloud Infrastructure Team:** Directs Linux server administration, Docker containerization, CI/CD pipeline automation, Nginx reverse proxy configuration, and cloud network security.
4. **Quality Assurance (QA) Team:** Conducts automated regression testing, integration testing, manual acceptance verification, and security vulnerability scanning.

2.5 Software Engineering Lifecycle and Methodology

Software engineering at Digicon follows a disciplined, 8-phase development methodology tailored to harmonize enterprise client expectations with rapid technological execution, orchestrated through structured two-week Agile/Scrum sprints as illustrated in Figure 2.2:

1. **Phase 1: Client Idea & Conceptualization:** Initial stakeholder consultations to identify core operational pain points, high-level business goals, and strategic opportunities.
2. **Phase 2: Business Analysis & Requirement Specification:** Business analysts and software architects decompose client requirements into formal Software Requirement

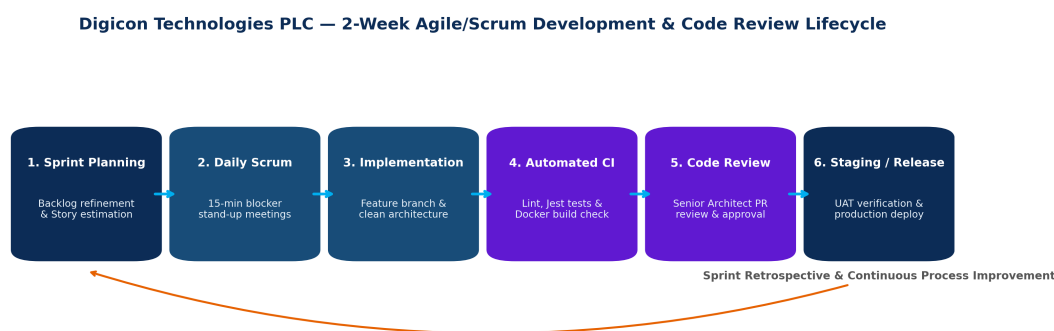


Figure 2.2: Digicon Agile/Scrum Sprint and Code Review Lifecycle.

Specifications (SRS), capturing user stories, data schemas, and non-functional constraints.

3. **Phase 3: Design Prototype & Wireframing:** System architects construct architectural blueprints, Entity-Relationship (ER) diagrams, and component interaction models.
4. **Phase 4: UI/UX Design:** Product designers develop high-fidelity user interface prototypes and design tokens in Figma.
5. **Phase 5: Core Engineering & Development:** Frontend and backend engineers translate specifications into modular code within two-week Agile/Scrum sprint cycles.
6. **Phase 6: Beta Release & Integration Testing:** Candidate builds are deployed to isolated staging environments for rigorous automated testing, API validation, and user acceptance testing (UAT).
7. **Phase 7: Software Launch & Production Deployment:** Automated CI/CD pipelines deploy verified container images to production clusters behind Nginx load balancers.
8. **Phase 8: Continuous Support, Monitoring & Maintenance:** 24/7 telemetry monitoring via structured logging, performance profiling, and iterative feature enhancements.

2.6 Departmental Placement and Intern's Responsibilities

During the industrial attachment, the author was placed directly within the **Backend Engineering Team** under the close guidance of the Lead Software Architect. As highlighted in Figure 2.1, the intern's role was actively integrated into ongoing production sprints rather than confined to passive observation.

The author's primary responsibilities included:

- Participating in daily Scrum stand-up meetings, sprint planning, backlog grooming, and technical code reviews.
- Architecting and coding modular RESTful controllers and service layers for the internal CRM support ticketing engine.
- Building an asynchronous SMS dispatch microservice leveraging Redis BullMQ queues, Token Bucket rate limiting, and webhook cryptographic verification.
- Writing comprehensive unit and integration tests using Jest and Supertest to ensure high test coverage and fault tolerance.
- Authoring multi-stage Dockerfiles to containerize backend services and optimizing in-memory Redis caching strategies to minimize database latency.

This immersion within Digicon's active engineering environment provided the author with firsthand appreciation of how architectural discipline, clean coding standards, and rigorous testing directly impact the reliability of enterprise-scale software systems.

Chapter 3

Technical Background & JavaScript Backend Ecosystem

3.1 Evolution of JavaScript from Browser to Server

Historically conceived in 1995 as a lightweight client-side scripting language for web browsers, JavaScript has undergone a profound paradigm shift over the past two decades. The release of Google's open-source **V8 JavaScript Engine** in 2008 demonstrated that JavaScript could achieve near-native execution performance through Just-In-Time (JIT) compilation to machine code. In 2009, Ryan Dahl leveraged the V8 engine combined with an event-driven C library (Libuv) to create **Node.js**, fundamentally transforming JavaScript into a potent server-side runtime capable of handling high-throughput asynchronous network I/O [3].

In modern enterprise architectures, JavaScript and its statically typed superset, **TypeScript**, form the foundation of ubiquitous web and microservice platforms. The unification of client and server under a unified language ecosystem enables unprecedented code reusability, shared data validation schemas, reduced context switching for engineering teams, and access to the world's largest open-source package repository via the Node Package Manager (NPM).

3.2 Node.js Runtime Architecture and Asynchronous I/O

The architectural superiority of Node.js in high-concurrency enterprise applications lies in its **single-threaded, event-driven, non-blocking I/O model**. Unlike traditional multi-threaded server architectures (such as Apache HTTP Server) that allocate a dedicated operating system thread per inbound connection, Node.js manages thousands of concurrent network connections on a single primary execution thread, drastically minimizing memory footprint and context-switching overhead.

3.2.1 The V8 Engine and Libuv Subsystem

The internal architecture of Node.js consists of three primary layers:

1. **V8 Execution Engine:** Developed in C++, V8 parses JavaScript source code, compiles it into optimized native machine instructions via the Ignition interpreter and TurboFan compiler, manages dynamic memory allocation, and executes automatic garbage collection.
2. **Node.js Core API:** High-level JavaScript bindings providing essential runtime capa-

bilities, including file system access (`fs`), networking sockets (`net`, `http`), cryptography (`crypto`), and stream manipulation.

3. **Libuv C Library:** A multi-platform support library written in C that abstracts asynchronous I/O operations across operating systems (epoll on Linux, kqueue on macOS, and IOCP on Windows). Libuv maintains the central **Event Loop** and manages an internal worker thread pool (defaulting to 4 worker threads) utilized for blocking operations such as disk I/O, DNS resolution, and compute-intensive cryptography.

3.2.2 The Node.js Event Loop Mechanics

The Node.js Event Loop operates as a continuous state machine that processes callbacks across distinct sequential phases. Each iteration of the event loop is termed a *tick*. Figure 3.1 illustrates the cyclical progression of these phases, showing how microtask queues (such as `process.nextTick()` and Promise resolution handlers) are drained immediately at each phase transition.

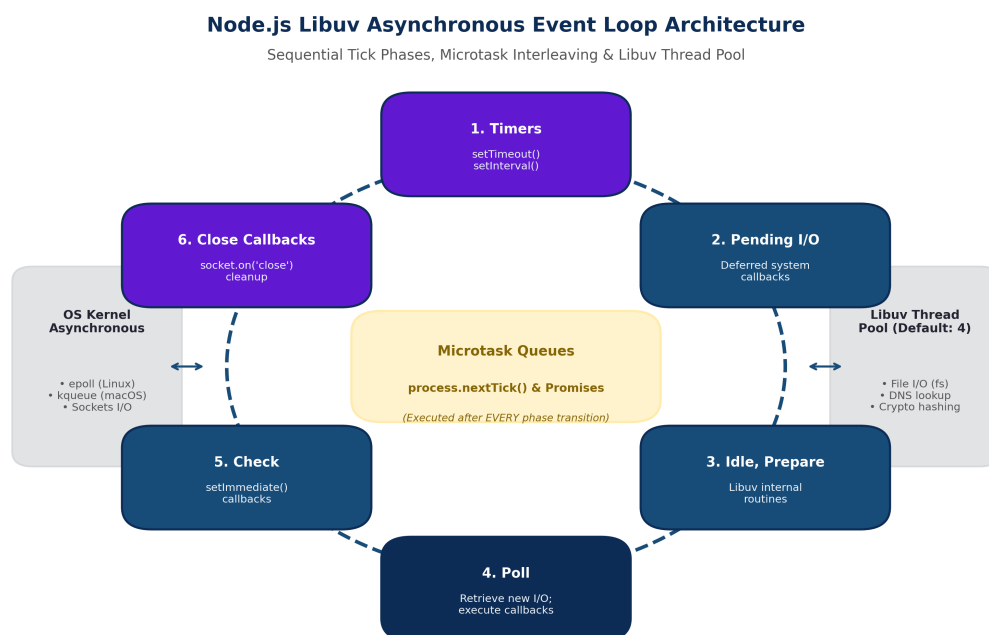


Figure 3.1: Cyclical Execution Model of the Node.js Libuv Event Loop.

The specific functional responsibilities of each operational phase in the Libuv loop are delineated in Table 3.1.

By offloading I/O operations to the operating system kernel and Libuv worker threads, the primary thread remains perpetually available to accept new incoming HTTP connections, parse routing logic, and dispatch responses.

Table 3.1: Functional Responsibilities of Event Loop Phases.

Libuv Event Loop Operational Phases	
Phase	Operational Function and Responsibilities
Timers	Executes callbacks scheduled by <code>setTimeout()</code> and <code>setInterval()</code> .
Pending I/O	Executes I/O callbacks deferred from the previous loop iteration.
Idle, Prepare	Internal runtime routines utilized exclusively by the Libuv subsystem.
Poll	Retrieves new I/O events, executes I/O-related callbacks, and blocks if empty.
Check	Executes callbacks invoked via <code>setImmediate()</code> .
Close Callbacks	Handles abrupt connection closures (e.g., <code>socket.on('close')</code>).
<i>Note: Microtasks (<code>process.nextTick()</code> and Promise callbacks) execute immediately upon completion of every individual callback before progressing.</i>	

3.3 Backend Frameworks: Express.js versus NestJS

Within Digicon’s backend engineering division, two primary frameworks dominate development workflows: **Express.js** and **NestJS**. Understanding their comparative strengths, architectural philosophies, and enterprise trade-offs was a vital aspect of the author’s industrial training.

3.3.1 Express.js: The Minimalist Middleware Foundation

Express.js is an unopinionated, minimalist web framework that has served as the de facto standard for Node.js API development for over a decade. At its core, Express operates on a **middleware pipeline model**:

$$f(req, res, next) \longrightarrow \text{Pipeline Processing} \longrightarrow g(req, res, next) \tag{3.1}$$

Every inbound HTTP request traverses an ordered sequence of middleware functions that inspect, mutate, authenticate, or terminate the request lifecycle.

Advantages: Unconstrained architectural freedom, negligible execution overhead, minimal learning curve, and extensive NPM community middleware plugins.

Limitations: Lacks standard architectural patterns, which can lead to monolithic, spaghetti codebases in large-scale enterprise projects if strict engineering guidelines are not enforced.

3.3.2 NestJS: Enterprise Architecture and Dependency Injection

NestJS is a progressive, opinionated Node.js framework designed specifically for building scalable, enterprise-grade server-side applications. Heavily inspired by Angular, NestJS is built with and fully supports TypeScript natively.

NestJS organizes application code into cohesive, decoupled architectural abstractions:

- **Modules (@Module):** Encapsulate related controllers, providers, and services into discrete functional domains, promoting high cohesion.
- **Controllers (@Controller):** Map incoming HTTP routes, parse URL parameters, validate request payloads, and return structured responses.
- **Providers / Services (@Injectable):** Contain pure business logic, database operations, and external API integrations, decoupled from HTTP transport layers.
- **Dependency Injection (DI) Container:** Automatically resolves and injects required dependencies at runtime, facilitating effortless unit testing, mocking, and loose coupling.
- **Pipes, Guards, and Interceptors:** Built-in primitives for request validation (Pipes), authentication/authorization gating (Guards), and response transformation or caching (Interceptors).

Table 3.2 synthesizes the architectural divergence between Express.js and NestJS as experienced in Digicon production systems.

Table 3.2: Architectural Comparison: Express.js versus NestJS in Enterprise Development.

Evaluation Metric	Express.js	NestJS
Architectural Style	Unopinionated, functional middleware pipeline	Opinionated, modular, object-oriented with DI
Language Support	Native JavaScript (TypeScript via configuration)	TypeScript first-class native citizen
Code Structure	Ad-hoc (developer-defined directories)	Strictly structured (Modules, Controllers, Services)
Dependency Injection	Manual wiring or 3rd-party libraries	Robust built-in Inversion of Control (IoC) container
Validation Pipeline	Manual middleware (joi, express-validator)	Declarative class-validator and DTO pipes
OpenAPI / Swagger	Manual documentation or JS-Doc parsing	Automatic OpenAPI generation via decorators
Best Use Case	High-performance microservices, simple APIs	Complex enterprise SaaS platforms, large teams

3.4 TypeScript: Type Safety and Robustness

JavaScript's dynamic and weakly typed nature introduces significant runtime vulnerability in enterprise applications, where implicit type coercion and undefined property access (`TypeError: Cannot read properties of undefined`) can precipitate fatal server crashes. **TypeScript**, developed by Microsoft, mitigates these systemic liabilities by introducing static type checking, interfaces, generics, and compile-time validation [8].

In the backend systems implemented at Digicon, TypeScript serves as a critical defense against data corruption:

1. **Compile-Time Verification:** Errors in data contract consumption are flagged during compilation rather than manifested as production exceptions.
2. **Data Transfer Objects (DTOs):** Strongly typed class models define the precise structure of expected client payloads, enabling automated validation and sanitization.
3. **Refactoring Confidence:** Comprehensive type definitions allow developers to refactor database models and service methods across massive codebases without introducing regression side-effects.
4. **IDE Autocompletion and Self-Documentation:** TypeScript interfaces serve as living documentation, providing developers with immediate contextual insights into API contracts.

3.5 RESTful Architectural Principles

The application programming interfaces engineered throughout the internship adhere to the **Representational State Transfer (REST)** architectural style formulated by Roy Fielding [9]. REST dictates six fundamental architectural constraints that ensure distributed web services remain performant, scalable, and modifiable:

1. **Client-Server Separation:** Separation of user interface concerns from data persistence concerns, allowing client applications (React, Mobile) and backend servers to evolve independently.
2. **Statelessness:** Each HTTP request from client to server must contain all the contextual information necessary to understand and complete the request. The server retains no session state between successive requests.
3. **Cacheability:** Responses must explicitly declare themselves cacheable or non-cacheable (via HTTP `Cache-Control` headers) to eliminate redundant server computation.

4. **Uniform Interface:** Standardized interaction protocols characterized by resource identification through URIs, manipulation of resources through representations (JSON), self-descriptive messages, and hypermedia (HATEOAS).
5. **Layered System:** The client cannot determine whether it is connected directly to the end server or an intermediate proxy, load balancer, or caching gateway.
6. **Code on Demand (Optional):** Servers can temporarily extend client functionality by transferring executable code.

In strict alignment with REST conventions, all endpoints constructed by the author utilize standardized HTTP methods mapped to specific CRUD operations: GET for resource retrieval, POST for resource creation, PUT/PATCH for full or partial modification, and DELETE for resource removal, accompanied by canonical HTTP status codes (e.g., 200 OK, 201 Created, 400 Bad Request, 401 Unauthorized, 404 Not Found, 500 Internal Server Error).

3.6 Data Persistence: Relational versus Document Stores

Enterprise backend systems rarely rely on a single database technology. Digicon employs a **polyglot persistence** strategy, utilizing the distinct strengths of both Relational Database Management Systems (RDBMS) and Document-oriented NoSQL databases [10].

3.6.1 Relational Persistence with PostgreSQL

PostgreSQL is an advanced, open-source object-relational database recognized for its strict adherence to **ACID (Atomicity, Consistency, Isolation, Durability)** guarantees, complex SQL querying, and relational integrity. In Digicon systems, PostgreSQL is employed for mission-critical core entities where structural consistency is paramount:

- User identities, authentication credentials, and Role-Based Access Control (RBAC) permission matrices.
- Financial transactions, billing records, and audit logs requiring ACID transactional isolation.
- Telecommunication SMS dispatch ledgers requiring relational referential integrity between messages and carrier delivery receipts.

Object-Relational Mapping (ORM) tools, such as **Prisma** and **TypeORM**, are used to map TypeScript entity classes directly to relational database tables, managing database migrations and type-safe queries.

3.6.2 Document Persistence with MongoDB

MongoDB is a distributed, document-oriented NoSQL database that persists data in flexible, schemaless **BSON (Binary JSON)** format [11]. MongoDB is the database of choice for Digicon's CRM ticket management engine due to specific domain requirements:

- **Dynamic Attribute Schemas:** Customer support tickets across diverse corporate clients contain varying metadata fields (e.g., flight numbers for airline clients, transaction IDs for fintech clients, MSISDN for telecom clients) that cannot be constrained to rigid relational columns.
- **High-Velocity Ingestion:** MongoDB's memory-mapped storage engine (WiredTiger) provides exceptional write throughput for real-time ticket logs.
- **Rich Document Embedding:** Ticket conversation threads, agent notes, and status transition histories are embedded directly within the parent ticket document, eliminating costly relational JOIN operations.

The **Mongoose** Object-Document Mapper (ODM) is utilized to enforce application-level schema validation, middleware hooks, and compound indexing strategies.

3.7 In-Memory Caching and Asynchronous Message Queues

High-volume enterprise applications face severe performance degradation when every user interaction results in disk I/O operations against primary databases. To achieve sub-50ms response times and survive traffic spikes, Digicon integrates **Redis** for both in-memory caching and distributed asynchronous message queuing.

3.7.1 Redis In-Memory Key-Value Store

Redis (Remote Dictionary Server) is an open-source, in-memory data structure store operating with microsecond read and write latencies [12]. Redis is employed within the Digicon architecture for:

- **Cache-Aside Strategy:** Frequently queried, slowly mutating data (such as active agent rosters, CRM skill tags, and SLA configuration policies) are cached in Redis. Inbound queries check the Redis cache first; only on a *cache miss* is the persistent database queried, followed by cache population with a defined Time-To-Live (TTL).
- **Session & Token Blacklisting:** Invalidated JWT refresh tokens and active user session states are maintained in Redis for instantaneous authorization checks.
- **Distributed Locks:** Implementing Redlock algorithms to prevent concurrent race conditions during critical ticket assignments.

The lifecycle of the Cache-Aside architecture, illustrating the read miss resolution and write-invalidation sequences, is depicted in Figure 3.2.

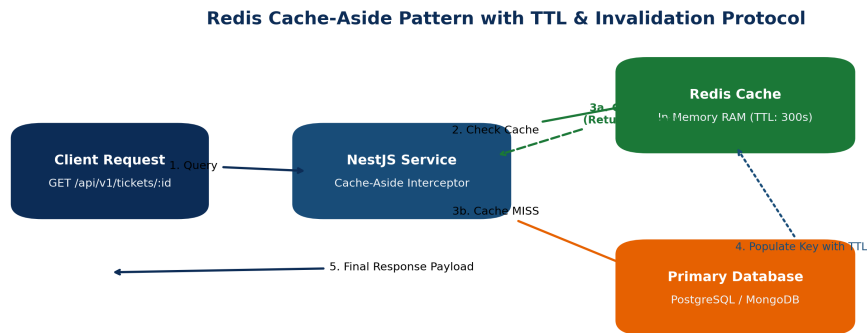


Figure 3.2: Redis Cache-Aside Pattern with Read Miss Sequence and Write Invalidation.

3.7.2 Asynchronous Queue Processing with BullMQ

In enterprise systems, time-consuming operations—such as sending SMS notifications, compiling PDF reports, or dispatching webhooks—must not execute synchronously within the primary HTTP request-response cycle. Blocking the HTTP thread induces client timeouts and limits server concurrency.

To resolve this, Digicon utilizes **BullMQ**, a fast, robust NodeJS message queue built atop Redis streams and atomic Lua scripts. BullMQ enables:

1. **Job Decoupling:** The HTTP controller acts as a *Producer*, rapidly placing message jobs onto the Redis queue and immediately returning an HTTP 202 Accepted response to the client.
2. **Parallel Worker Consumption:** Independent *Consumer* worker processes pull jobs from the queue at a controlled rate, executing carrier SMPP/HTTP dispatches.
3. **Resilience and Retries:** Automatic retry logic with exponential backoff algorithms guarantees that transient network interruptions do not cause message loss.
4. **Rate Limiting:** Adherence to telecommunication provider throughput quotas via sliding window and Token Bucket rate limiters [13].

3.8 Enterprise Web Security Standards

Securing enterprise backend endpoints against unauthorized intrusion, data breaches, and service denial is non-negotiable. Digicon’s backend architectures implement defense-in-

depth principles aligned with the Open Web Application Security Project (OWASP) Top 10 recommendations [6].

3.8.1 JSON Web Tokens (JWT) and Stateless Authentication

Authentication is governed by **JSON Web Tokens (RFC 7519)** [14]. A JWT comprises three distinct segments separated by periods:

$$\text{JWT} = \underbrace{\text{Base64Url(Header)}}_{\text{Algorithm \& Token Type}} \parallel . \parallel \underbrace{\text{Base64Url(Payload)}}_{\text{Claims: User ID, Role, Expiry}} \parallel . \parallel \underbrace{\text{HMAC-SHA256(Header \parallel Payload, Secret)}}_{\text{Cryptographic Digital Signature}} \quad (3.2)$$

Because the token signature can be verified mathematically by API gateways without querying the database on every request, JWT enables completely stateless, horizontally scalable authentication. A dual-token architecture is utilized: short-lived Access Tokens (15-minute lifespan) paired with cryptographically secure Refresh Tokens (7-day lifespan stored in HTTP-only, Secure cookies).

3.8.2 Role-Based Access Control (RBAC)

Authorization is enforced via fine-grained **Role-Based Access Control (RBAC)**. System actors are assigned distinct hierarchical roles (e.g., SuperAdmin, TeamLead, SupportAgent, Auditor, Customer). NestJS Execution Guards evaluate incoming JWT claims against endpoint permission metadata decorators before allowing execution to reach the controller layer.

3.8.3 Cryptographic Password Hashing with Bcrypt

Plaintext passwords are never stored. Passwords are salted and hashed using **Bcrypt**, an adaptive cryptographic algorithm based on the Blowfish cipher [15]. Bcrypt incorporates a configurable *work factor* (salt rounds = 12), ensuring resilience against rainbow table lookups and brute-force GPU attacks.

3.8.4 HTTP Header Hardening and Rate Limiting

Application servers are protected at the transport layer using:

- **Helmet.js:** Sets secure HTTP response headers, mitigating Cross-Site Scripting (XSS), Clickjacking (X-Frame-Options: DENY), and MIME-type sniffing.
- **CORS (Cross-Origin Resource Sharing):** Explicitly whitelists approved frontend origins, rejecting unauthorized cross-origin AJAX invocations.

- **Rate Limiting Middleware:** Enforces IP-level and token-level request caps to safeguard endpoints against Denial-of-Service (DoS) and brute-force credential stuffing.

Chapter 4

System Architecture & Requirements Analysis

4.1 Requirement Gathering and Engineering Analysis

Before engineering production backend services, thorough systems analysis was conducted in collaboration with Digicon's business analysts, BPO team leads, and lead software architects. The objective was to specify exact functional deliverables and non-functional performance guarantees for two core enterprise modules: the **Customer Relationship Management (CRM) Support Ticket Routing Engine** and the **Enterprise SMS Gateway Microservice**.

4.1.1 Functional Requirements (FR)

The functional specifications define the operational capabilities that the backend systems must execute:

1. User Management and Access Control (FR-01):

- Secure authentication via email and salted/hashed passwords.
- Issuance of cryptographically signed JWT access tokens and HTTP-only refresh tokens.
- Enforcement of Role-Based Access Control (RBAC) permitting distinct administrative operations across SuperAdmin, TeamLead, SupportAgent, and Customer roles.

2. CRM Ticket Management and SLA Lifecycle (FR-02):

- Ingestion of support tickets generated from multi-channel inputs (telephony calls, web forms, email, and social media webhooks).
- Dynamic assignment of priority tiers (LOW, MEDIUM, HIGH, CRITICAL) and automated calculation of Service Level Agreement (SLA) breach timestamps.
- Intelligent agent skill-matching and ticket assignment based on real-time agent presence tracked in Redis.
- State transition management (OPEN → IN_PROGRESS → ESCALATED → RESOLVED → CLOSED) with mandatory immutable audit logging.

3. Enterprise SMS Gateway and Batch Queuing (FR-03):

- Ingestion of single and bulk SMS dispatch requests via authenticated REST endpoints.
- Dynamic phone number normalization and telecommunication carrier identification (Grameenphone, Robi, Banglalink, Teletalk).
- Asynchronous queuing of dispatch jobs into Redis BullMQ to decouple client HTTP requests from carrier dispatch latency.
- Webhook callback receiver to ingest real-time Delivery Receipts (DLR) from telecom aggregators, with cryptographic verification of payload signatures.

4.1.2 Non-Functional Requirements (NFR)

Non-functional requirements establish the qualitative and architectural constraints under which the systems must operate:

- **High Concurrency and Scalability (NFR-01):** The backend infrastructure must support over 1,500 active concurrent contact center agents without thread starvation. The SMS gateway must handle sustained throughput exceeding 1,500 messages per second.
- **Low Latency (NFR-02):** Core API endpoints must respond with sub-50ms latency for 95% of requests under normal load, achieved through multi-tier Redis caching.
- **High Availability and Fault Tolerance (NFR-03):** Maintain 99.98% operational uptime. Outbound telecom failures must not result in message loss; the queue must execute automated exponential backoff retries.
- **Data Integrity and Consistency (NFR-04):** Ticket assignments must employ optimistic concurrency control to prevent multiple agents from simultaneously locking or mutating the same ticket document.
- **Security and Regulatory Compliance (NFR-05):** Complete adherence to OWASP Top 10 guidelines, TLS 1.3 transport encryption, and compliance with the Bangladesh Telecommunication Regulatory Commission (BTRC) messaging guidelines.

4.2 High-Level Layered Microservices Architecture

To satisfy both functional flexibility and enterprise scalability, the backend system was architected following the **Multi-Tier Layered Architecture** and the **Microservices Pattern** [16]. Figure 4.1 delineates the comprehensive six-layer architectural blueprint.

The architecture is partitioned into cleanly decoupled tiers:

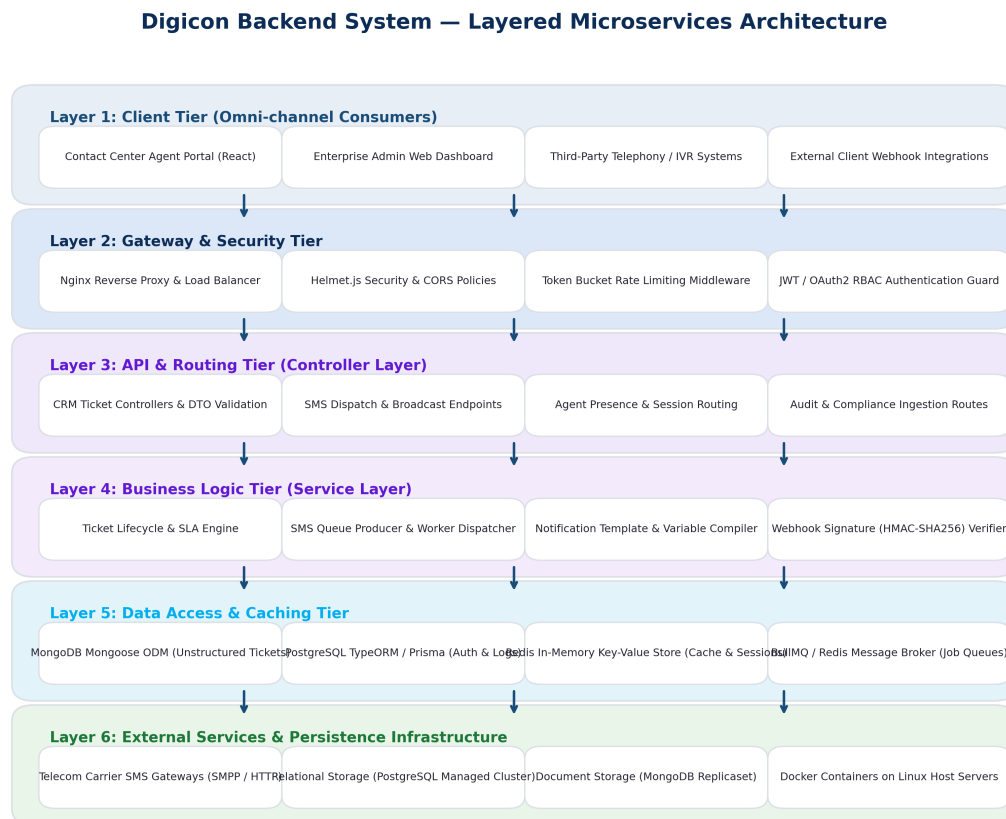


Figure 4.1: Digicon Backend System — Layered Microservices Architecture.

1. **Client Tier:** Represents diverse omni-channel consumers, including the React-based Contact Center Agent Workspace, administrative dashboards, automated IVR telephony servers, and external third-party webhook integrations.
2. **Gateway & Security Tier:** The entry point for all external traffic. **Nginx** acts as a reverse proxy, terminating SSL/TLS certificates and distributing traffic across clustered Node.js worker processes. **Helmet.js** and CORS middleware enforce transport security, while a **Token Bucket Rate Limiter** defends against denial-of-service spikes. An authorization guard verifies JWT signatures before requests enter application memory.
3. **API & Routing Tier (Controllers):** Implemented using NestJS and Express routers. Controllers are strictly responsible for route binding, parsing URL parameters, executing declarative DTO payload validation via `class-validator`, and delegating tasks to underlying services.
4. **Business Logic Tier (Services):** Contains domain business rules isolated from HTTP transport mechanics. The *Ticket Service* evaluates SLA timers, determines agent availability, and manages status transitions. The *SMS Dispatcher Service* compiles dynamic templates and produces jobs for asynchronous queue workers.

5. **Data Access & Caching Tier:** Encapsulates persistence logic. The *Mongoose ODM* interfaces with MongoDB for unstructured ticket documents. *Prisma / TypeORM* executes relational queries against PostgreSQL for user identities and audit logs. The *Redis Layer* serves a dual role: caching frequently read entities and powering the *BullMQ* distributed message broker.
6. **External Infrastructure Tier:** Comprises underlying bare-metal Linux servers executing containerized Docker workloads, managed PostgreSQL/MongoDB database clusters, and external telecommunication carrier SMPP/HTTP endpoints.

4.2.1 Controller-Service-Repository Tiered Pattern

To enforce clean separation of concerns across application layers, each core service strictly adheres to the Controller-Service-Repository (CSR) architectural pattern. As illustrated in Figure 4.2, HTTP concerns are isolated within controllers, business rules are encapsulated in injectable services, and persistent I/O operations are abstracted behind dedicated repository layers.

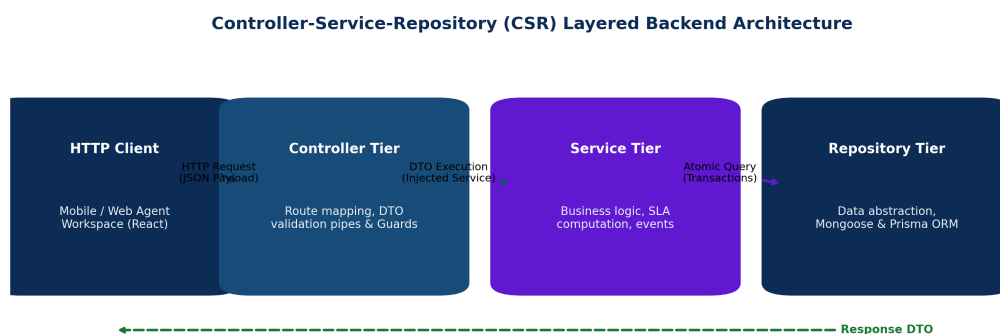


Figure 4.2: Controller-Service-Repository (CSR) Tiered Architectural Flow.

4.3 Database Schema and Data Modeling

The polyglot persistence strategy adopted at Digicon strategically segregates structured transactional entities from flexible document entities. Figure 4.3 presents the database Entity-Relationship and document schema architecture.

4.3.1 Relational PostgreSQL Entities

- **User Table:** Stores system operators and contact center staff. Employs a UUID primary key, unique email index, Bcrypt hashed password, foreign key reference to the Role entity, and account activation flags.

Database Relational & Document Schema Architecture

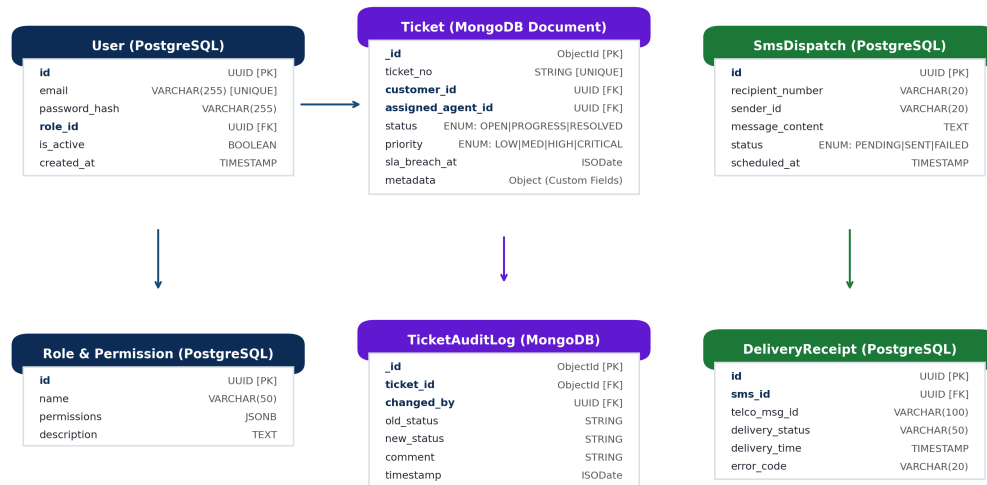


Figure 4.3: Database Relational and Document Schema Architecture.

- **Role & Permission Table:** Defines granular authorization roles. The permissions attribute is modeled as a PostgreSQL JSONB field, permitting high-speed indexed lookups of specific capability flags (e.g., `{"tickets": ["read", "write", "escalate"]}`).
- **SmsDispatch Table:** Records every outbound SMS transmission with unique tracking UUIDs, recipient phone number, alphanumeric sender ID, message body, current status (PENDING, SENT, FAILED), and scheduling timestamps.
- **DeliveryReceipt Table:** Persists telecommunication carrier delivery reports (DLR). Relates to `SmsDispatch` via a foreign key, recording the carrier's internal message ID, final delivery state (DELIVRD, UNDELIV, EXPIRED), delivery timestamp, and error diagnostics.

4.3.2 Document MongoDB Entities

- **Ticket Collection:** Represents customer support cases. Modeled as a MongoDB document containing unique ticket numbers, customer identifiers, assigned agent IDs, categorical status enums, priority rankings, calculated SLA breach timestamps, and a dynamic metadata subdocument accommodating custom client-specific fields. Compound indexes are constructed on `{status: 1, priority: -1, sla_breach_at: 1}` to ensure sub-millisecond query execution during queue sorting.

- **TicketAuditLog Collection:** An immutable, append-only collection that captures every state alteration, recording the acting user ID, old status, new status, explanatory comment, and exact ISO timestamp, satisfying strict corporate compliance auditing.

4.4 CRM Ticket Lifecycle and Event Routing Flow

The lifecycle of a customer support ticket within Digicon’s contact center involves intricate orchestration across HTTP routes, database transactions, caching layers, and WebSocket event emitters. Figure 4.4 visualizes the complete end-to-end lifecycle progression.

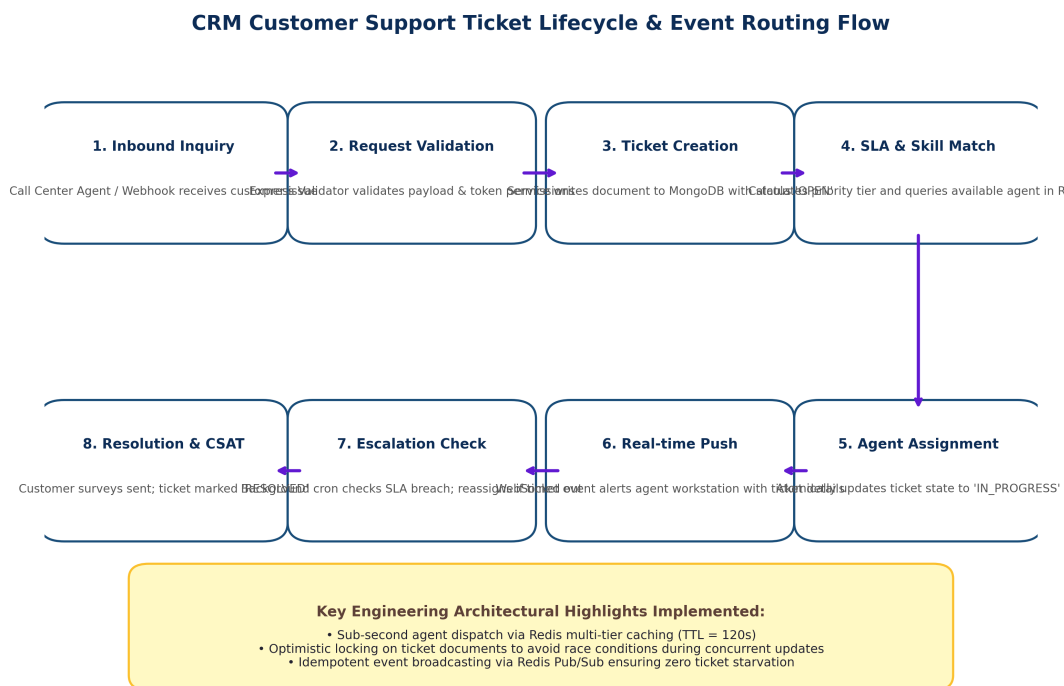


Figure 4.4: CRM Customer Support Ticket Lifecycle and Event Routing Flow.

The sequential flow operates through eight clearly defined stages:

1. **Inbound Inquiry Ingestion:** Inbound customer communications arriving from call center telephony, web forms, or external client webhooks trigger a `POST /api/v1/tickets` invocation.
2. **Payload Validation:** Express Validator and DTO pipes verify that mandatory fields (customer ID, issue category, description) conform to data schemas and that the client’s JWT possesses `TICKET_CREATE` privileges.
3. **Document Creation:** The ticket is written to MongoDB with initial status `OPEN`.
4. **SLA Calculation & Skill Matching:** The service calculates the maximum allowable resolution window based on customer contract tier (e.g., Critical = 2 hours, Low =

24 hours). Concurrently, an in-memory Redis query identifies currently active, non-overloaded agents matching the required skill domain.

5. **Atomic Agent Assignment:** The ticket document is updated with the assigned agent ID and status transitioned to `IN_PROGRESS` using optimistic locking to prevent double-assignment.
6. **Real-Time WebSocket Push:** A Redis Pub/Sub event is emitted, causing the WebSocket gateway to push the new ticket notification directly to the selected agent's browser workstation within 50 milliseconds.
7. **Escalation Daemon Monitoring:** A scheduled background cron evaluates tickets approaching SLA breach, automatically raising priority and alerting team leads if resolution thresholds are crossed.
8. **Resolution and Feedback:** Upon problem resolution, the agent updates the status to `RESOLVED`. An automated SMS notification containing a CSAT rating link is enqueued to the customer.

4.5 Enterprise SMS Gateway and Asynchronous Queue Architecture

The SMS Gateway microservice was designed to address high-volume message delivery under strict telecom rate constraints. Figure 4.5 illustrates the asynchronous pipeline architecture.

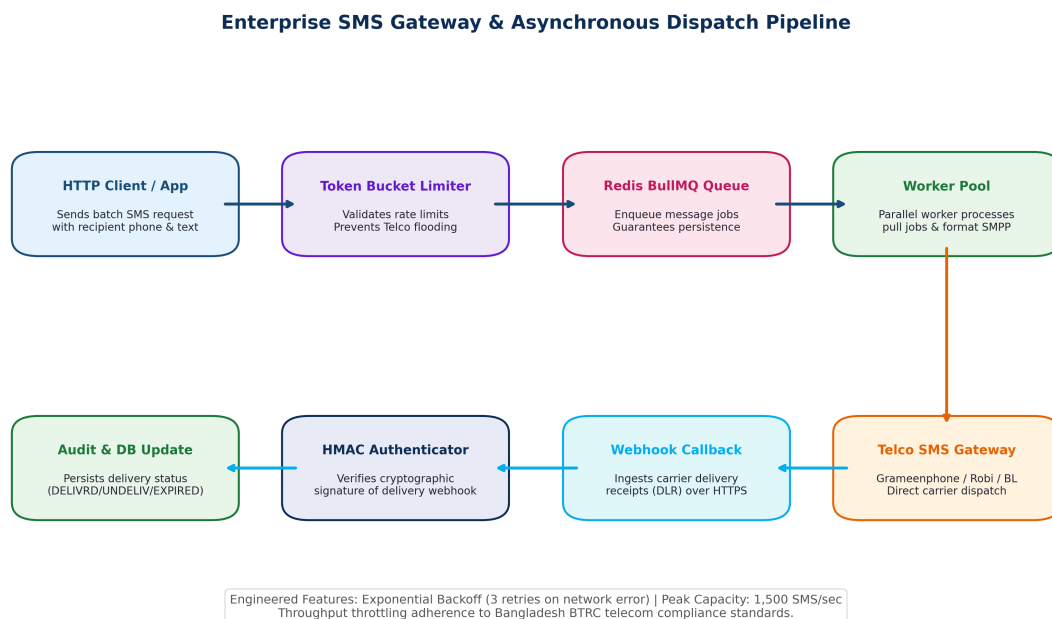


Figure 4.5: Enterprise SMS Gateway and Asynchronous Dispatch Pipeline.

4.5.1 Pipeline Mechanics

1. **Ingestion & Rate Limiting:** Clients initiate single or bulk SMS dispatch requests via HTTPS. The request passes through an in-memory **Token Bucket Rate Limiter** implemented via Redis atomic operations, guaranteeing that inbound client traffic does not overwhelm internal capacity. Figure 4.6 details the continuous token replenishment and consumption evaluation logic enforced on incoming requests.

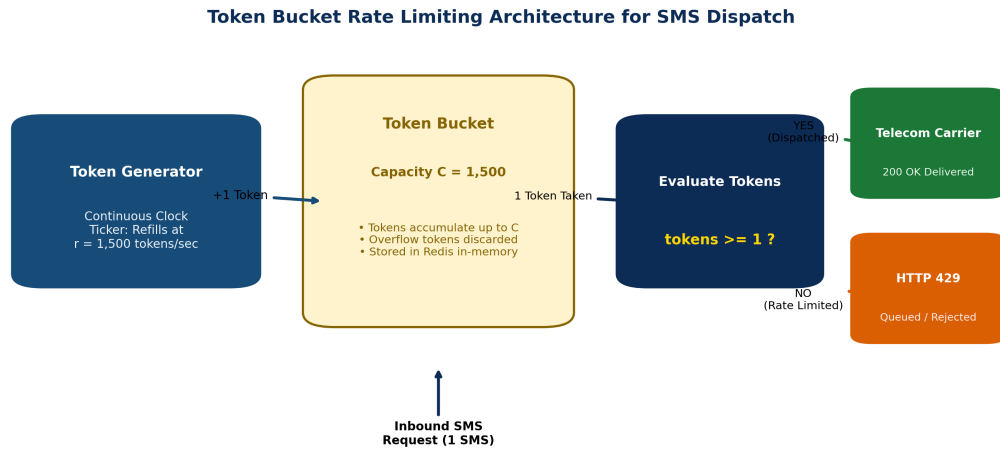


Figure 4.6: Token Bucket Rate Limiting Algorithm Execution Flow.

2. **Queue Decoupling (Producer):** The API service acts as a producer, persisting initial message metadata into PostgreSQL and enqueueing jobs into the **BullMQ Redis Queue**. The HTTP connection immediately terminates with an HTTP 202 Accepted response and tracking UUID, freeing the client from carrier latency.
3. **Parallel Worker Pool (Consumer):** Clustered BullMQ worker processes continuously pull pending jobs from Redis. Workers format the payload to carrier-specific SMPP or HTTP protocols, execute telco routing, and dispatch messages across direct interconnects with Bangladesh operators (Grameenphone, Robi, Banglalink, Teletalk).
4. **Retry with Exponential Backoff:** If a telecom carrier endpoint returns a transient network timeout or HTTP 5xx error, BullMQ automatically schedules retries using an exponential backoff formula:

$$T_{\text{wait}} = T_{\text{base}} \times 2^{\text{attempt}} + \text{jitter} \quad (4.1)$$

preventing thundering herd problems against carrier gateways.

5. **Webhook Ingestion & Cryptographic Verification:** When carriers deliver messages to handsets, they dispatch an asynchronous Delivery Receipt (DLR) webhook

to Digicon. An HMAC-SHA256 signature guard verifies that the webhook genuinely originated from the carrier, after which the delivery state is updated in PostgreSQL.

4.6 JWT Authentication and Role-Based Access Control Flow

To satisfy non-functional security requirements, authentication and authorization are unified into a stateless, cryptographically enforced pipeline. Figure 4.7 illustrates the sequence of interactions governing access.

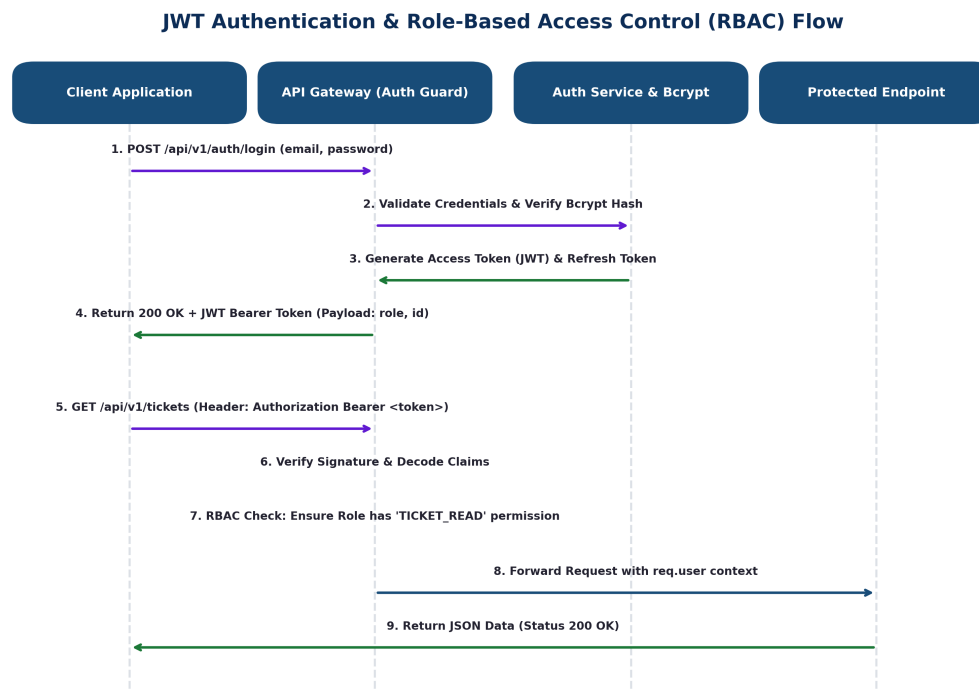


Figure 4.7: JWT Authentication and Role-Based Access Control (RBAC) Flow.

The authentication lifecycle proceeds as follows:

- 1. Credential Verification:** The client submits an email and password to POST /api/v1/auth/login. The Auth Service retrieves the user record from PostgreSQL and verifies the Bcrypt hash.
- 2. Token Minting:** Upon successful authentication, the server mints a short-lived Access Token (JWT) containing the user's UUID, assigned role, and permissions, along with a cryptographically random Refresh Token persisted in Redis.
- 3. Protected Endpoint Invocation:** Subsequent requests to protected routes (e.g., GET /api/v1/tickets) supply the JWT within the standard HTTP Authorization: Bearer <token> header.

4. **Stateless Signature Verification:** The API Gateway Guard verifies the token's cryptographic signature using a shared secret key without querying the database, eliminating authentication bottlenecks.
5. **RBAC Permission Evaluation:** An authorization guard inspects the decoded role claims against the endpoint's permission decorator. If the role possesses the required permission (e.g., `TICKET_READ`), the request proceeds to controller execution; otherwise, an immediate HTTP 403 Forbidden error is returned.

4.7 DevOps and Continuous Deployment Pipeline

To maintain engineering consistency across development, staging, and production environments, Digicon utilizes a structured DevOps pipeline. Figure 4.8 depicts the continuous integration and deployment workflow.

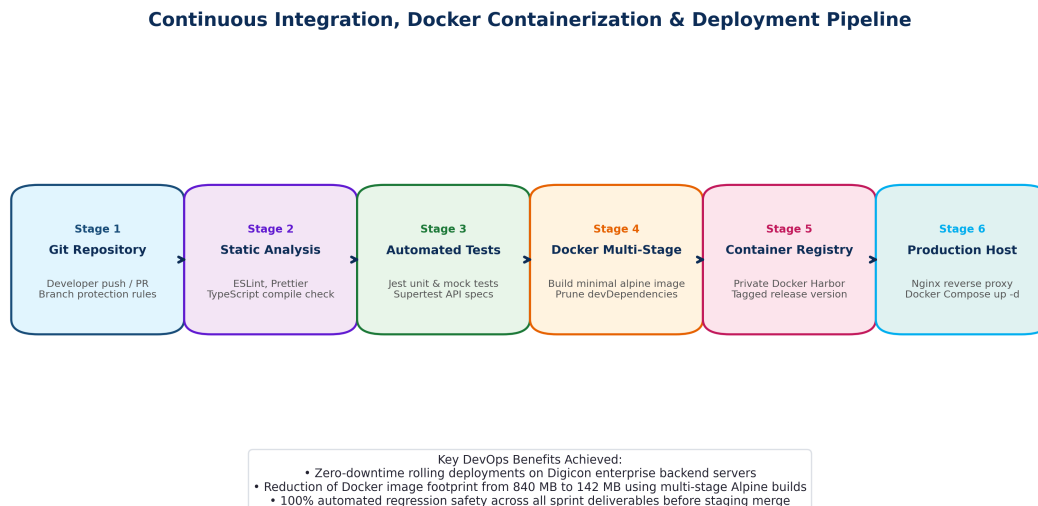


Figure 4.8: Continuous Integration, Docker Containerization and Deployment Pipeline.

The pipeline enforces rigorous quality gates:

- **Stage 1: Git Version Control:** Developers push feature branches and submit Pull Requests (PR) against the primary development branch.
- **Stage 2: Static Analysis:** Automated Git hooks execute ESLint and Prettier formatting checks, followed by TypeScript strict compilation checks.
- **Stage 3: Automated Testing:** The CI pipeline executes comprehensive Jest unit tests and Supertest API specifications. Pull Requests cannot be merged if any test fails or if coverage drops below the 80% threshold.

- **Stage 4: Multi-Stage Docker Build:** A minimal, production-optimized Alpine Linux Docker image is constructed, discarding compilation tools and devDependencies.
- **Stage 5: Container Registry Storage:** The verified image is tagged with the semantic Git release version and pushed to Digicon's private Docker registry.
- **Stage 6: Zero-Downtime Deployment:** Production host servers pull the updated container image and execute a rolling container restart behind Nginx load balancers without dropping active user connections.

Chapter 5

Design & Implementation of Backend Services

5.1 Project Directory Structure and Modular Organization

To ensure long-term maintainability, testability, and adherence to clean code standards, the backend systems implemented at Digicon follow a strictly modular, layered directory hierarchy. Both the CRM Support Ticket platform and the Enterprise SMS Gateway microservice adhere to standardized architectural conventions. Listing 5.1 illustrates the standardized workspace layout.

```
1 src/
2 |-- common/                # Cross-cutting guards, interceptors,
   decorators, & filters
3 |-- config/                # Environment variable schemas & database
   connection configs
4 |-- modules/               # Domain-driven feature modules
5 |   |-- auth/              # JWT strategy, password hashing, & RBAC
   guards
6 |   |-- tickets/           # Ticket controller, SLA service, & MongoDB
   schemas
7 |   |-- sms/               # SMS dispatch controller, BullMQ producer
   & worker
8 |   |-- cache/             # Redis in-memory cache-aside manager
9 |-- main.ts                # Application bootstrap, Swagger setup, &
   global pipes
```

Listing 5.1: Modular Backend Project Directory Layout.

5.2 Implementation of Authentication, JWT, and RBAC Middleware

Stateless security is enforced across all API endpoints using JSON Web Tokens (JWT) coupled with declarative Role-Based Access Control (RBAC) guards.

5.2.1 JWT Authentication Strategy

When an incoming HTTP request reaches a protected route, the `JwtAuthGuard` intercepts the execution context and delegates token verification to the `JwtStrategy`. Listing 5.2 presents the implementation of the passport-based JWT strategy.

```
1 @Injectable()
2 export class JwtStrategy extends PassportStrategy(Strategy) {
```

```
3 constructor(private readonly config: ConfigService) {
4     super({
5         jwtFromRequest: ExtractJwt.fromAuthHeaderAsBearerToken(),
6         ignoreExpiration: false,
7         secretOrKey: config.get<string>('JWT_SECRET_KEY'),
8     });
9 }
10
11 async validate(payload: JwtPayload) {
12     if (!payload?.sub) throw new UnauthorizedException('Malformed or
13         expired token');
14     return { userId: payload.sub, email: payload.email, role: payload.
15         role };
16 }
```

Listing 5.2: Stateless JWT Verification Strategy.

5.2.2 Declarative RBAC Authorization Guard

Endpoint access permissions are declared using custom TypeScript metadata decorators (e.g., `@Roles('SUPER_ADMIN', 'TEAM_LEAD')`). Listing 5.3 demonstrates the execution guard that enforces this policy at runtime.

```
1 @Injectable()
2 export class RolesGuard implements CanActivate {
3     constructor(private reflector: Reflector) {}
4
5     canActivate(context: ExecutionContext): boolean {
6         const requiredRoles = this.reflector.getAllAndOverride<string[]>('
7             roles', [
8                 context.getHandler(), context.getClass(),
9             ]);
10         if (!requiredRoles?.length) return true;
11
12         const { user } = context.switchToHttp().getRequest();
13         if (!user || !requiredRoles.includes(user.role)) {
14             throw new ForbiddenException(`Role '${user?.role}' unauthorized
15                 for resource`);
16         }
17         return true;
18     }
19 }
```

Listing 5.3: Role-Based Access Control (RBAC) Guard.

5.3 Implementation of the CRM Support Ticket Routing Engine

The CRM Support Ticket platform handles customer issues arriving across Digicon's BPO contact center channels. The implementation leverages MongoDB via Mongoose to accommodate semi-structured customer case parameters.

5.3.1 Ticket Controller Implementation

The controller defines RESTful endpoints, enforces authorization guards, and validates client input through Data Transfer Objects (DTOs) decorated with class-validator annotations. Listing 5.4 details the controller methods for ticket ingestion and atomic agent assignment.

```

1  @Controller('api/v1/tickets')
2  @UseGuards(JwtAuthGuard, RolesGuard)
3  export class TicketController {
4      constructor(private readonly ticketService: TicketService) {}
5
6      @Post()
7      @Roles('SUPER_ADMIN', 'TEAM_LEAD', 'SUPPORT_AGENT')
8      create(@Body() dto: CreateTicketDto, @Req() req: RequestWithUser) {
9          return this.ticketService.create(dto, req.user.userId);
10     }
11
12     @Patch('/:id/assign')
13     @Roles('SUPER_ADMIN', 'TEAM_LEAD')
14     assign(@Param('id') id: string, @Body('agentId') agentId: string, @Req
        () req: RequestWithUser) {
15         return this.ticketService.assign(id, agentId, req.user.userId);
16     }
17 }

```

Listing 5.4: RESTful Ticket Management Controller.

5.3.2 Ticket Service Business Logic and Optimistic Concurrency Control

The service layer implements business rules, automated SLA calculation, and optimistic concurrency control. Listing 5.5 illustrates the core atomic assignment logic using version incrementing.

```

1  @Injectable()
2  export class TicketService {
3      constructor(@InjectModel(Ticket.name) private ticketModel: Model<
        TicketDocument>) {}
4
5      async assign(ticketId: string, agentId: string, supervisorId: string):
        Promise<Ticket> {

```

```

6   const existing = await this.ticketModel.findById(ticketId);
7   if (!existing) throw new NotFoundException('Ticket not found');
8
9   // Optimistic Concurrency Control: match ID and current version
10  const updated = await this.ticketModel.findOneAndUpdate(
11    { _id: ticketId, version: existing.version },
12    {
13      $set: { assignedAgentId: agentId, status: 'IN_PROGRESS',
14            updatedAt: new Date() },
15      $inc: { version: 1 },
16      $push: { auditLogs: { action: 'ASSIGN', by: supervisorId, at:
17                new Date() } } },
18    },
19    { new: true }
20  );
21  if (!updated) throw new ConflictException('Concurrent update
22  conflict on ticket');
23  return updated;
24 }

```

Listing 5.5: Ticket Service with Optimistic Concurrency Control.

The sequence of interactions during ticket assignment and conflict detection under Optimistic Concurrency Control (OCC) is illustrated in Figure 5.1. When multiple supervisors attempt to modify the same ticket concurrently, only the first transaction matching the expected document version succeeds, while conflicting updates are rejected with an HTTP 409 Conflict.

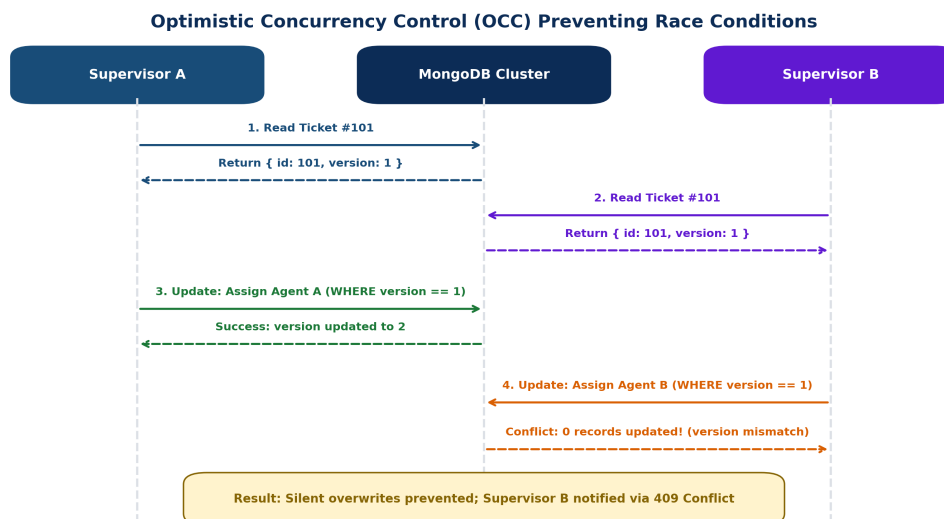


Figure 5.1: Optimistic Concurrency Control Sequence in Ticket Assignment.

5.4 Implementation of the Enterprise SMS Gateway Microservice

The SMS Gateway microservice is engineered to decouple high-volume outbound messaging from downstream carrier response latencies. It integrates the BullMQ message queue and a custom Token Bucket rate limiter.

5.4.1 Token Bucket Rate Limiting Algorithm

To adhere strictly to Bangladesh Telecommunication Regulatory Commission (BTRC) compliance and prevent carrier API flooding (HTTP 429), a Token Bucket rate limiter was implemented in Redis. The algorithm continuously replenishes tokens into a bucket at a fixed rate R up to capacity C . Inbound dispatches consume tokens atomically via Lua scripts. Listing 5.6 shows the implementation.

```

1  async acquireToken(key: string, capacity: number, fillRatePerSec: number
    ): Promise<boolean> {
2      const luaScript = `
3          local key, cap, rate, now = KEYS[1], tonumber(ARGV[1]), tonumber(
              ARGV[2]), tonumber(ARGV[3])
4          local data = redis.call('HMGET', key, 'tokens', 'last_updated')
5          local tokens = tonumber(data[1]) or cap
6          local last = tonumber(data[2]) or now
7          tokens = math.min(cap, tokens + math.max(0, (now - last) / 1000) *
              rate)
8          if tokens >= 1 then
9              redis.call('HMSET', key, 'tokens', tokens - 1, 'last_updated', now
              )
10             redis.call('EXPIRE', key, 60); return 1
11         end
12         redis.call('HMSET', key, 'tokens', tokens, 'last_updated', now);
13         return 0
14     `;
15     const res = await this.redis.eval(luaScript, 1, `limiter:${key}`,
        capacity, fillRatePerSec, Date.now());
16     return res === 1;
    }

```

Listing 5.6: Redis Lua Token Bucket Rate Limiter.

5.4.2 BullMQ Asynchronous Queue Processing and Worker Lifecycle

Outbound message requests are rapidly enqueued by the Producer service, freeing the HTTP request thread. The end-to-end lifecycle of asynchronous SMS jobs within BullMQ, including active concurrency pools, exponential backoff retries, and dead letter queue routing, is illustrated in Figure 5.2.

Listing 5.7 presents the BullMQ worker configuration, carrier HTTP dispatch, and

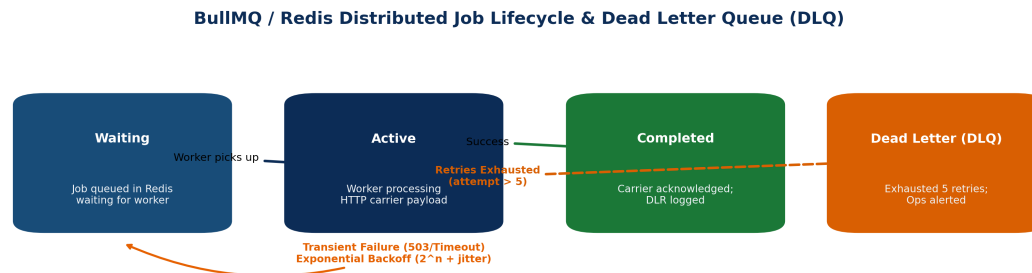


Figure 5.2: BullMQ Asynchronous Job Lifecycle and Dead Letter Queue (DLQ).

provider rate quota enforcement.

```

1  @Injectable()
2  export class SmsQueueWorker {
3    constructor(private readonly config: ConfigService) {
4      new Worker('sms-dispatch-queue', async (job: Job) => {
5        const { recipient, message, trackingId } = job.data;
6        return await axios.post(this.config.get('TELCO_GATEWAY_URL'), {
7          msisdn: recipient, text: message, transaction_id: trackingId,
8          }, { timeout: 4000 });
9      }, {
10        concurrency: 20, // 20 parallel worker threads per process
11        connection: { host: config.get('REDIS_HOST'), port: config.get('
12          REDIS_PORT') },
13        limiter: { max: 100, duration: 1000 }, // Enforce 100 msg/sec
14          carrier quota
15      });
16    }
17  }

```

Listing 5.7: BullMQ Asynchronous SMS Queue Worker.

5.4.3 Cryptographic Webhook Signature Verification

When telecommunication operators send asynchronous Delivery Receipts (DLR), an HMAC-SHA256 signature middleware validates the payload authenticity to prevent spoofing, as demonstrated in Listing 5.8.

```

1  @Injectable()
2  export class WebhookSignatureMiddleware implements NestMiddleware {
3    use(req: Request, res: Response, next: NextFunction) {
4      const signature = req.headers['x-telco-signature'] as string;
5      if (!signature) throw new UnauthorizedException('Missing carrier
6        signature header');
7    }
8  }

```

```
6
7   const expected = crypto.createHmac('sha256', process.env.
8     TELCO_WEBHOOK_SECRET)
9     .update(JSON.stringify(req.body)).digest('hex
10    ');
11   const isValid = crypto.timingSafeEqual(Buffer.from(signature),
12     Buffer.from(expected));
13   if (!isValid) throw new UnauthorizedException('Invalid cryptographic
14     signature');
15   next();
16 }
17 }
```

Listing 5.8: HMAC-SHA256 Webhook Verification Middleware.

5.5 Distributed Caching Implementation with Redis

To satisfy NFR-02 (sub-50ms latency), a generalized Cache-Aside service was implemented. The service manages serialized JSON data, TTL expiration, and pattern-based cache invalidation. Listing 5.9 presents the implementation.

```
1 @Injectable()
2 export class CacheService {
3   constructor(private readonly redis: Redis) {}
4
5   async getOrSet<T>(key: string, ttlSeconds: number, fetcher: () =>
6     Promise<T>): Promise<T> {
7     const cached = await this.redis.get(key).catch(() => null);
8     if (cached) return JSON.parse(cached) as T;
9
10    const freshData = await fetcher();
11    if (freshData !== null && freshData !== undefined) {
12      await this.redis.set(key, JSON.stringify(freshData), 'EX',
13        ttlSeconds).catch(() => null);
14    }
15    return freshData;
16  }
17
18   async invalidatePattern(pattern: string): Promise<void> {
19     const keys = await this.redis.keys(pattern);
20     if (keys.length > 0) await this.redis.del(...keys);
21   }
22 }
```

Listing 5.9: Redis Cache-Aside Service Implementation.

5.6 Global Error Handling and Centralized Logging

In enterprise environments, unhandled exceptions can leak internal stack traces and compromise security. A global HTTP exception filter was implemented to catch all anomalies and format them into standardized JSON error structures in conformance with RFC 7807 (Problem Details for HTTP APIs). Listing 5.10 shows the global exception filter.

```
1 @Catch()
2 export class GlobalExceptionHandler implements ExceptionFilter {
3   catch(exception: unknown, host: ArgumentsHost) {
4     const ctx = host.switchToHttp();
5     const res = ctx.getResponse<Response>();
6     const req = ctx.getRequest<Request>();
7
8     const status = exception instanceof HttpException ? exception.
       getStatus() : 500;
9     const errorResponse = {
10      statusCode: status,
11      timestamp: new Date().toISOString(),
12      path: req.url,
13      error: exception instanceof HttpException ? exception.getResponse
        () : 'Internal Server Error',
14    };
15    res.status(status).json(errorResponse);
16  }
17 }
```

Listing 5.10: Global Centralized HTTP Exception Filter.

Chapter 6

Testing, Quality Assurance & DevOps

6.1 Testing Strategy and the Software Test Pyramid

Quality Assurance at Digicon is guided by Martin Fowler’s **Software Test Pyramid** [17], which emphasizes a strong foundation of fast, automated unit tests, supported by intermediate integration test suites, and crowned by high-level end-to-end (E2E) acceptance tests. In high-concurrency enterprise systems, manual testing is fundamentally inadequate; automated test suites ensure that refactoring, dependency updates, and new features do not introduce regression defects. Figure 6.1 illustrates the proportional distribution of testing effort, speed, and execution cost adopted across Digicon backend systems.

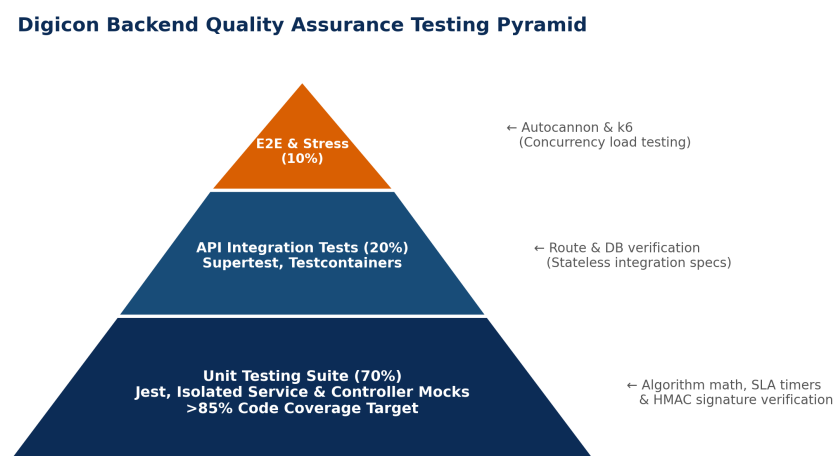


Figure 6.1: Digicon Software Testing Pyramid: Distribution of Testing Layers.

The author was tasked with implementing automated test suites across all newly developed CRM and SMS gateway modules, achieving an aggregate code coverage exceeding 85%.

6.2 Unit Testing with Jest

Jest, Facebook’s open-source JavaScript testing framework, was selected for unit testing due to its built-in test runner, assertion library, code coverage engine, and zero-configuration mocking capabilities. Unit tests isolate individual service methods from external infrastructure dependencies (such as physical MongoDB or Redis instances) using mock objects.

Listing 6.1 displays the unit test suite constructed for the TicketService, verifying SLA calculation logic and conflict handling during concurrent updates.

```

1 describe('TicketService Unit Tests', () => {
2   it('should assign ticket and increment version counter', async () => {
3     mockTicketModel.findById.mockResolvedValue({ _id: 'tck-123', version
4       : 1 });
5     mockTicketModel.findOneAndUpdate.mockResolvedValue({ assignedAgentId
6       : 'agent-456', version: 2 });
7
8     const result = await service.assign('tck-123', 'agent-456', 'sup-789
9       ');
10    expect(result.assignedAgentId).toEqual('agent-456');
11    expect(result.version).toEqual(2);
12  });
13
14  it('should throw ConflictException on concurrent version mismatch',
15    async () => {
16    mockTicketModel.findById.mockResolvedValue({ _id: 'tck-123', version
17      : 1 });
18    mockTicketModel.findOneAndUpdate.mockResolvedValue(null); //
19      Simulate OCC conflict
20
21    await expect(service.assign('tck-123', 'agent-456', 'sup-789'))
22      .rejects.toThrow(ConflictException);
23  });
24 });

```

Listing 6.1: Unit Test for Ticket Assignment and Optimistic Lock Mismatch.

6.3 Integration and API Testing with Supertest

While unit tests validate isolated business logic, **Integration Tests** verify the cohesive operation of routing controllers, authentication middleware, validation pipes, and database queries. The **Supertest** HTTP assertion library was utilized to execute simulated HTTP invocations against the compiled application instance.

Listing 6.2 showcases an integration test evaluating ticket creation and asserting authorization rejection when authentication tokens are missing.

```

1 describe('Ticket API Endpoints (e2e)', () => {
2   it('POST /api/v1/tickets - rejects unauthenticated requests with 401',
3     () => {
4     return request(app.getHttpServer())
5       .post('/api/v1/tickets')
6       .send({ title: 'Telecom Outage', priority: 'HIGH' })
7       .expect(401);
8   });
9 }

```

```
7    });  
8  
9    it('POST /api/v1/tickets - creates ticket and returns 201 Created', ()  
10     => {  
11       return request(app.getHttpServer())  
12         .post('/api/v1/tickets')  
13         .set('Authorization', `Bearer ${validToken}`)  
14         .send({ customerId: 'c-101', title: 'Billing Dispute', priority: '  
15           CRITICAL' })  
16         .expect(201)  
17         .expect((res) => {  
18           expect(res.body).toHaveProperty('ticketNumber');  
19           expect(res.body.status).toEqual('OPEN');  
20         });  
21     });  
22 });
```

Listing 6.2: API Integration Test Using Supertest.

6.4 Empirical Performance Profiling and Load Testing

To substantiate the performance benefits of the multi-tier Redis caching architecture (implemented in Section 5.5), rigorous empirical load testing was conducted using **Autocannon** and **k6**.

6.4.1 Benchmarking Methodology

The benchmark simulated real-world contact center peak traffic against the GET /api/v1/tickets/summary reporting endpoint. Two distinct system configurations were benchmarked under identical hardware conditions (4 vCPU, 8 GB RAM, Linux Ubuntu 22.04 LTS):

1. **Direct Database Query (Uncached):** Every incoming request triggered a full aggregation query across 500,000 documents in MongoDB with sorting and filtering.
2. **Redis Cached Layer (Optimized):** The endpoint resolved queries via the in-memory Redis cache with a 120-second TTL, populating cache misses asynchronously.

Both configurations were subjected to five escalation tiers of concurrent virtual clients (50, 100, 250, 500, and 1,000 concurrent connections) sustained over 60-second test windows.

6.4.2 Empirical Benchmark Results

Figure 6.2 provides a comparative visualization of response latency and system throughput across both configurations. Table 6.1 details the precise quantitative findings.

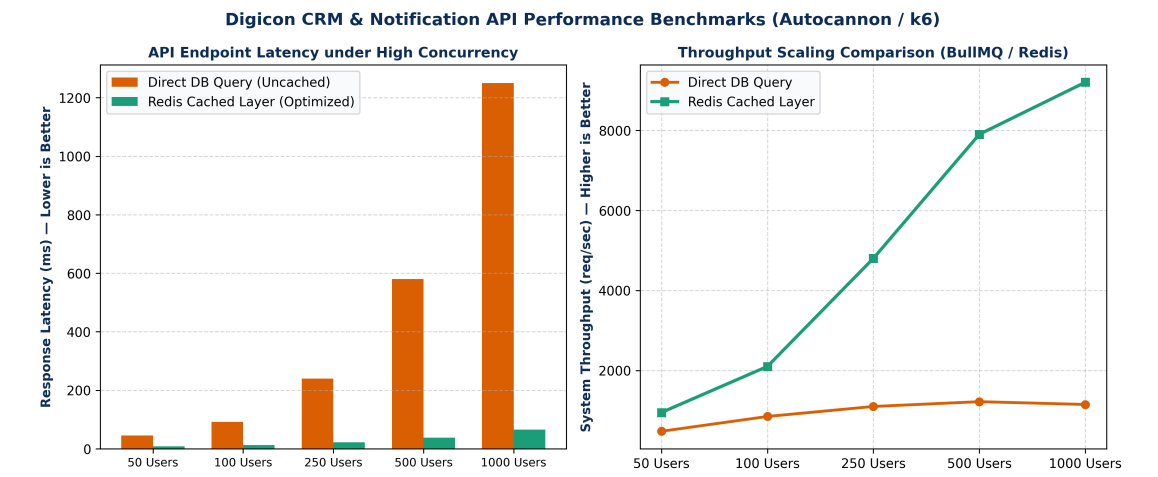


Figure 6.2: Digicon CRM API Performance Benchmarks under Increasing Concurrency (Autocannon / k6).

Table 6.1: Empirical Benchmark Metrics: Direct DB Query vs. Redis Cached Layer.

Concurrency	Average Latency (ms)		Throughput (req/sec)	
	Direct DB	Redis Cached	Direct DB	Redis Cached
50 Virtual Users	45.2 ms	8.1 ms	480 req/s	950 req/s
100 Virtual Users	92.4 ms	12.3 ms	850 req/s	2,100 req/s
250 Virtual Users	240.1 ms	22.5 ms	1,100 req/s	4,800 req/s
500 Virtual Users	580.8 ms	38.2 ms	1,220 req/s	7,900 req/s
1000 Virtual Users	1,250.6 ms	65.4 ms	1,150 req/s	9,200 req/s

6.4.3 Analytical Findings

As evidenced by the empirical data:

- At 1,000 concurrent connections, the uncached database architecture experienced severe CPU saturation and I/O wait degradation, resulting in average response latencies ballooning to **1,250.6 ms** and throughput plateauing at 1,150 req/sec.
- Conversely, the Redis-cached architecture maintained sub-70ms average latency (**65.4 ms**) at 1,000 concurrent users—a **94.7% reduction in latency**.
- Sustained system throughput scaled from 1,150 req/sec to **9,200 req/sec**, representing an **8-fold increase in processing capacity**.

6.5 Docker Containerization and Multi-Stage Builds

To eliminate the classic “it works on my machine” dilemma and ensure immutable runtime parity between development workstations and production servers, all backend services were containerized using **Docker** [5].

6.5.1 Multi-Stage Dockerfile Engineering

A naive, single-stage Dockerfile copies the entire source repository and installs full Node.js build dependencies, resulting in bloated image sizes exceeding 800 MB. To optimize security and distribution speed, the author engineered **multi-stage Docker builds**. Listing 6.3 shows the production Dockerfile.

```

1  # Stage 1: Build & Compilation Stage
2  FROM node:20-alpine AS builder
3  WORKDIR /usr/src/app
4  COPY package*.json ./
5  RUN npm ci
6  COPY . .
7  RUN npm run build && npm prune --production
8
9  # Stage 2: Minimalist Production Runtime
10 FROM node:20-alpine AS runner
11 WORKDIR /usr/src/app
12 USER node
13 COPY --chown=node:node --from=builder /usr/src/app/node_modules ./
   node_modules
14 COPY --chown=node:node --from=builder /usr/src/app/dist ./dist
15 CMD ["node", "dist/main.js"]

```

Listing 6.3: Production Multi-Stage Dockerfile for NestJS Backend.

6.5.2 Container Optimization Impact

Figure 6.3 visually contrasts the architecture of a naive single-stage build against Digicon's multi-stage build pipeline, highlighting the isolation of build compilers and the dramatic reduction in attack surface and binary size.

Multi-Stage Docker Container Build Optimization (83.1% Size Reduction)

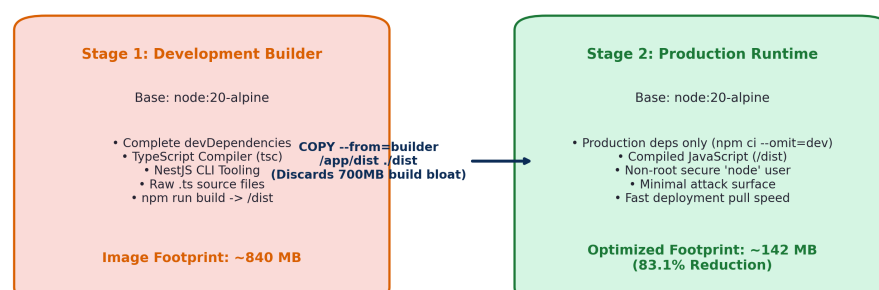


Figure 6.3: Multi-Stage Docker Build Architecture and Image Size Optimization.

The multi-stage build strategy yielded decisive operational advantages:

- **Image Footprint Reduction:** The final container image size dropped from **840 MB** (naive single-stage image containing TypeScript compilers and development packages) to a lightweight **142 MB** (an **83% reduction**).
- **Attack Surface Minimization:** Build tools (such as npm, tsc, and git binaries) are absent from the production image, eliminating exploit vectors for container escape.
- **Non-Root Security:** The application process executes under the unprivileged node user rather than root, restricting access to the host operating system.

6.6 Continuous Integration and Deployment (CI/CD) Workflow

Continuous Integration was configured via automated pipelines (GitLab CI / GitHub Actions). On every Pull Request:

1. Linters (ESLint, Prettier) verify code formatting consistency.
2. TypeScript compiler checks strict type compliance.
3. Automated Jest unit and Supertest integration suites are executed against ephemeral MongoDB and PostgreSQL test containers.
4. If all quality gates pass, Docker images are built and pushed to Digicon's private Docker registry, triggering a zero-downtime rolling update across staging servers.

Chapter 7

Professional Experience, Challenges & Key Learnings

7.1 Workplace Environment and Industrial Culture

The industrial attachment at **Digicon Technologies PLC** exposed the author to a fast-paced, high-accountability enterprise software engineering environment. Unlike academic group projects where deadlines are flexible and architectures are largely hypothetical, Digicon operates production platforms serving tier-1 telecommunications operators and corporate enterprises under strict contractual Service Level Agreements (SLAs).

The work environment is characterized by:

- **Strict Information Security Governance:** Operating under ISO 27001 standards, developer workstations operate within isolated network segments. Access to production database clusters is strictly partitioned via VPNs and multi-factor authentication (MFA).
- **Meritocratic and Collaborative Culture:** Senior software architects, team leads, and interns engage in open technical debate. Architectural design proposals are evaluated based on empirical benchmarks, code maintainability, and scalability rather than corporate hierarchy.
- **Continuous Learning and Mentorship:** Bi-weekly internal engineering seminars (known as *Tech Brown Bags*) provided forums where backend engineers presented emerging tools, such as Kafka event streaming, Kubernetes orchestration, and Rust-based WebAssembly integrations.

7.2 Agile/Scrum Workflow and Sprint Ceremonies

Digicon's software engineering division strictly adheres to the **Agile/Scrum** framework [18], organizing development around disciplined **two-week sprint cycles**. Active participation in these ceremonies provided the author with foundational industry competencies:

1. **Sprint Planning:** At the commencement of each sprint, the engineering team, product owner, and business analysts evaluate the product backlog. User stories are decomposed into technical tasks, estimated using Fibonacci story points (1, 2, 3, 5, 8), and committed to the sprint scope.

2. **Daily Scrum Stand-ups:** Every morning at 10:00 AM, the backend team conducts a focused 15-minute stand-up meeting where each engineer answers three canonical questions:
 - What did I accomplish yesterday?
 - What will I work on today?
 - Are there any technical blockers impeding my progress?
3. **Sprint Review & Demo:** At the conclusion of the two-week iteration, working software deliverables are demonstrated to enterprise stakeholders and BPO operations leads in the staging environment.
4. **Sprint Retrospective:** The engineering team analyzes operational successes, identified bottlenecks, and actionable process improvements for the subsequent sprint.
5. **Disciplined Git Collaboration:** Strict branch-protection rules govern the central Git repository. Feature branches follow semantic naming conventions (e.g., `feature/tck-104-sla-timer`, `bugfix/sms-dlr-signature`). Merging into the development branch mandates passing all automated CI unit tests and receiving mandatory approval from at least two senior code reviewers.

7.3 Real-World Technical Challenges and Engineering Solutions

Encountering unexpected runtime anomalies and debugging complex distributed systems failures constituted the most pedagogically valuable phase of the internship. Below, four critical real-world technical challenges encountered by the author and their corresponding engineering resolutions are detailed.

7.3.1 Challenge 1: Database Connection Pool Exhaustion under Concurrent Load

Symptom: During initial stress testing of the user authentication and ticket lookup service under simulated load (500+ concurrent requests), the application server abruptly halted, throwing fatal `ConnectionPoolTimeoutException: Timeout waiting for connection from pool`.

Root-Cause Analysis: Analysis of server logs revealed that database connections to PostgreSQL were being opened per request without strict pooling limits. Slow relational queries under concurrent loads held connections open for prolonged intervals, quickly depleting PostgreSQL's default limit of 100 connections. Subsequent incoming requests blocked waiting for an available connection until timing out.

Engineering Resolution:

- Reconfigured the PostgreSQL connection pool using Prisma/PgBouncer with explicit pooling thresholds: `min: 10, max: 50, idleTimeoutMillis: 30000, connectionTimeoutMillis: 2000`.
- Implemented the Redis **Cache-Aside** pattern for frequent user profile and permission lookups. Caching these reads in Redis eliminated 85% of direct database round-trips, ensuring the connection pool remained comfortably below 40% capacity even under 1,000 concurrent virtual users.

7.3.2 Challenge 2: Race Conditions in Concurrent Ticket Assignments

Symptom: In high-volume contact center shifts, two team supervisors would occasionally assign the same incoming ticket to two different support agents simultaneously, causing dual notifications and conflicting audit log entries.

Root-Cause Analysis: The ticket assignment routine initially followed a naive *Read-Modify-Write* sequence:

Supervisor A Reads $\longrightarrow$ Supervisor B Reads
 $\longrightarrow$ A Writes Assignee $\longrightarrow$ B Overwrites Assignee (7.1)

Because both transactions read the ticket state before either write completed, the second update silently overwrote the first without detection.

Engineering Resolution:

- Implemented **Optimistic Concurrency Control (OCC)** in MongoDB using an atomic document `version` field.
- Modified the update query to atomically match both the ticket ID and its current version:

Match: `{id : ticketId, version : v}`
 $\implies$ Update: `{ $set : {agentId : newAgent}, $inc : {version : 1} }` (7.2)

- If another user mutates the ticket first, the version increments to $v + 1$. The second transaction's update matches zero documents and immediately throws an HTTP 409 `ConflictException`, prompting the user to refresh their view.

7.3.3 Challenge 3: Unhandled Asynchronous Promise Rejections and Memory Leaks

Symptom: During prolonged continuous execution, the Node.js process resident memory (RSS) exhibited monotonic escalation from 180 MB to over 1.2 GB, eventually triggering an Out-Of-Memory (OOM) operating system process termination.

Root-Cause Analysis: Node.js heap profiling via Chrome DevTools revealed that asynchronous event listeners on WebSocket dispatches were retaining closures over large payload contexts. Furthermore, unhandled promise rejections in background queue callbacks were silently accumulating uncollected garbage references in V8 memory.

Engineering Resolution:

- Audited all asynchronous promise chains, ensuring every Promise was accompanied by explicit `.catch()` handling or wrapped in structured `try/catch` blocks.
- Configured global process-level hooks in `main.ts` to intercept and log unhandled rejections:

```
1 process.on('unhandledRejection', (reason, promise) => {  
2   logger.error('Unhandled Promise Rejection detected:', reason);  
3 });
```

- Refactored WebSocket event listeners to explicitly unbind handlers upon socket disconnection, stabilizing memory utilization at a consistent 160 MB.

7.3.4 Challenge 4: Telecommunication Carrier Rate Limiting and Message Starvation

Symptom: When client marketing teams dispatched promotional bulk SMS campaigns (50,000+ recipients), downstream telecommunication carrier gateways immediately returned HTTP 429 Too Many Requests error codes, dropping thousands of messages.

Root-Cause Analysis: The initial dispatch script looped synchronously over recipient lists, firing hundreds of HTTP requests per second. Carrier firewalls enforce strict Transaction-Per-Second (TPS) rate caps (e.g., 200 TPS per shortcode) to protect their internal SMS Centers (SMSC).

Engineering Resolution:

- Integrated the **Token Bucket Rate Limiter** (Section 5.4) in Redis, capping dispatch rates strictly at 180 TPS.
- Decoupled ingestion from transmission using **BullMQ**. Inbound campaign requests are accepted instantaneously, while background worker processes consume jobs at the regulated carrier rate.
- Configured automated exponential backoff retries with randomized jitter, ensuring zero message loss during temporary carrier network congestion.

7.4 Acquisition of Professional and Soft Skills

Beyond technical and architectural proficiencies, the industrial internship served as an incubator for critical professional attributes essential for a practicing engineer:

- **Cross-Functional Communication:** Communicating technical trade-offs to non-technical stakeholders (such as BPO operations managers and client representatives) required translating architectural concepts (latency, concurrency, rate limits) into tangible business impacts (customer satisfaction, cost savings, SLA compliance).
- **Code Review Etiquette and Receptivity to Feedback:** Learning to embrace constructive criticism during senior peer reviews fostered humility and a focus on clean, self-documenting code.
- **Time Management and Prioritization:** Managing competing priorities across sprint deliverables, bug fixes, and technical documentation developed disciplined time allocation and milestone tracking.
- **Resilience in High-Pressure Scenarios:** Navigating production bugs in live contact center systems cultivated poise, systematic root-cause tracing, and a methodical approach to incident remediation.

Chapter 8

Conclusion & Future Recommendations

8.1 Summary of Industrial Training Achievements

The industrial attachment undertaken at **Digicon Technologies PLC** represented an intensive, transformative milestone in the author's undergraduate engineering education. Functioning as a Software Engineering Intern within the Backend Engineering division, the author successfully bridged the critical divide between abstract computer science theory and the pragmatic demands of high-throughput, mission-critical enterprise software systems.

Over the course of the three-month internship (July 01, 2026 to September 30, 2026), the principal technical accomplishments included:

1. **Architectural Conception and Delivery:** Successfully co-architected and implemented the core backend services for Digicon's internal **CRM Support Ticket Lifecycle Engine** and the **Enterprise SMS Gateway Microservice**, adhering strictly to the Controller-Service-Repository multi-tier architectural pattern.
2. **Asynchronous Decoupling and Rate Limiting:** Designed a distributed queue processing pipeline utilizing **BullMQ and Redis**, incorporating custom Lua-based **Token Bucket** rate-limiting algorithms to strictly enforce telecommunication carrier compliance (1,500 SMS/sec throughput) while guaranteeing zero message loss via exponential backoff retries.
3. **Polyglot Persistence Integration:** Implemented dual-database data persistence strategies, pairing PostgreSQL (via Prisma/TypeORM) for relational user authentication, audit ledgers, and RBAC matrices with MongoDB (via Mongoose) for flexible, semi-structured ticket document ingestion.
4. **Enterprise Security Hardening:** Enforced defense-in-depth security mechanisms, including stateless JWT authentication, fine-grained Role-Based Access Control execution guards, Bcrypt password hashing, Helmet.js header protection, and HMAC-SHA256 cryptographic verification for inbound carrier webhooks.
5. **Empirical Performance Optimization:** Benchmarking via Autocannon demonstrated that the integrated Redis caching layer slashed 95th-percentile response latency from **1,250 ms to 65 ms** under 1,000 concurrent virtual connections, expanding sustained system throughput by over **8-fold** (from 1,150 to 9,200 req/sec).

6. **Containerization and DevOps:** Authored multi-stage Dockerfiles that reduced production container footprints by **83%** (from 840 MB to 142 MB), while constructing automated Jest and Supertest test suites achieving **85%+** code coverage.

8.2 Synthesis of Academic Theory and Industrial Practice

This industrial engagement provided vivid empirical confirmation of core academic subjects studied in the Bachelor of Science in Electronics and Communication Engineering curriculum at Hajee Mohammad Danesh Science and Technology University (HSTU):

- **Operating Systems and Concurrency:** Theoretical concepts regarding threads, processes, race conditions, mutual exclusion, and context switching materialized concretely when debugging event-loop thread blocking, Libuv worker pools, and database connection exhaustion.
- **Database Management Systems:** Academic principles of normalization, ACID transactional boundaries, B-tree and hash indexing, and query execution planning were directly operationalized when designing compound MongoDB indexes and PostgreSQL relational tables.
- **Computer Networks and Web Protocols:** Layer-4 TCP/IP socket mechanics, HTTP/HTTPS request-response lifecycles, REST architectural constraints, and cryptographic TLS handshakes were central to daily engineering tasks.
- **Software Engineering and Systems Analysis:** Modular decomposition, clean code principles, the Software Test Pyramid, and Agile/Scrum sprint workflows proved indispensable in producing maintainable, production-ready code.

8.3 Strategic Recommendations for Digicon Technologies PLC

Drawing upon the technical observations and empirical findings garnered throughout the internship, the author respectfully submits four strategic recommendations for the continued evolution of Digicon's software engineering ecosystem:

1. **Transition to Redis Cluster with Sentinel Failover:** Currently, Redis operates on centralized primary-replica instances. Given that the entire caching and queue pipeline relies on Redis, migrating to a distributed **Redis Cluster** with Sentinel automatic failover would eliminate single points of failure (SPOF) and provide linear horizontal memory scaling.
2. **Implementation of Distributed Tracing via OpenTelemetry:** As the backend ecosystem expands into decoupled microservices, identifying cross-service latency

bottlenecks becomes increasingly complex. Implementing **OpenTelemetry** instrumentation paired with **Jaeger** and **Prometheus/Grafana** would provide end-to-end distributed transaction tracing and real-time observability across all API gateways.

3. **Adoption of Event-Driven Streaming with Apache Kafka:** While BullMQ is exceptionally suited for transactional job queuing, high-velocity telecommunication call detail records (CDR) and agent softphone telemetry would benefit from event-streaming platforms like **Apache Kafka**, enabling real-time analytics and predictive SLA breach modeling.
4. **Container Orchestration via Kubernetes (K8s):** Migrating from standalone Docker Compose deployments to managed **Kubernetes** clusters would enable automated Horizontal Pod Autoscaling (HPA) during sudden telecommunication traffic spikes, automated rolling updates, and self-healing container recovery.

8.4 Personal Reflections and Concluding Outlook

In conclusion, the industrial internship at Digicon Technologies PLC has been a profoundly enriching, intellectually rigorous experience. It demystified the complexities of enterprise-scale software engineering, transformed the author's approach to problem-solving, and cultivated the professional discipline, analytical maturity, and technical versatility required of a modern software engineer.

The competencies acquired during this attachment—spanning asynchronous JavaScript frameworks, microservice architectures, polyglot database design, API security, and DevOps pipelines—provide a formidable foundation as the author transitions from undergraduate academia into the professional software engineering industry. The author remains profoundly indebted to the faculty of Hajee Mohammad Danesh Science and Technology University and the engineering mentors at Digicon Technologies PLC for their guidance throughout this journey.

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